



Giới thiệu công nghệ giải trình tự thế hệ mới (NGS) và ứng dụng trong lâm sàng

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Viện Nghiên cứu Ứng dụng Khoa học Sức khỏe và Lão hóa – Bệnh viện Thống Nhất

Nội dung

1. Giới thiệu dự án **BỘ GEN NGƯỜI & GIẢI TRÌNH TỰ GEN THỂ HỆ MỚI (NGS)**
2. Ví dụ mối quan hệ của **BIẾN THỂ GEN** và **BỆNH DI TRUYỀN**
3. Quy trình **XÉT NGHIỆM** gen bằng phương pháp giải trình tự thể hệ mới (NGS)
4. Năm ví dụ về **XÉT NGHIỆM** gen cho **BỆNH DI TRUYỀN** và **UNG THƯ**
5. Ứng dụng NGS trong bệnh nhiễm



Xin gửi lời cảm ơn trân thành đến



Bệnh viện Thống Nhất và Viện ARIHA

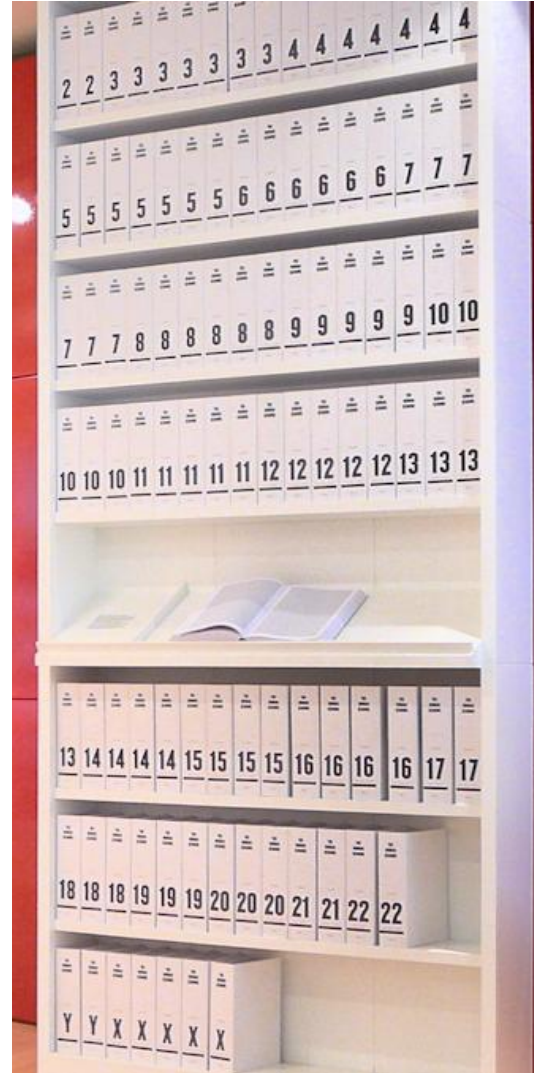
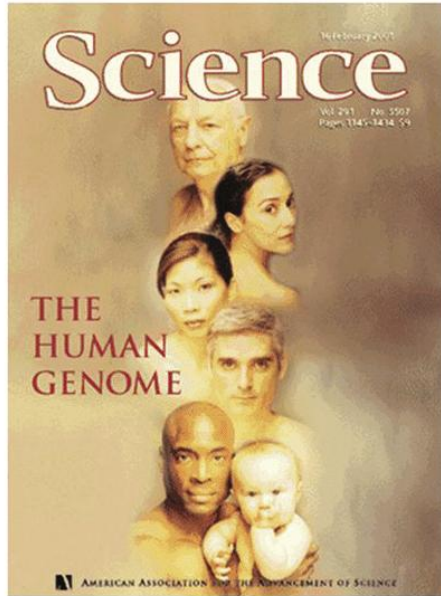
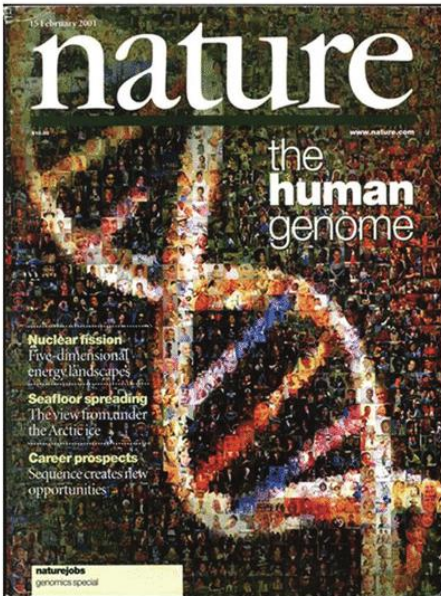
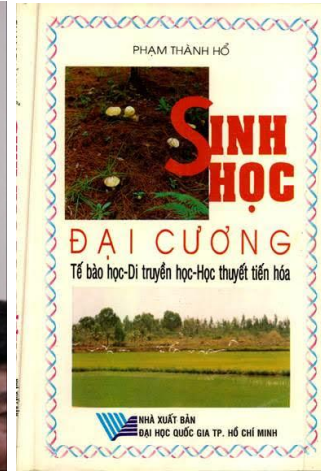
- **PGS. TS. Lê Đình Thanh**
- **PGS. TS. Võ Thành Toàn**
- **ThS. Nguyễn Văn Tài**
- **ThS. Nguyễn Thị Duyên Anh**
- **BS. CK1 Trần Vũ Minh Tâm**
- **ThS. Cao Thị Hoài**
- **ThS. DS Lê Đặng Minh Anh**
- **CN. Trần Hoàng Uyên**
- **CN. Phạm Hiếu Đan**
- **CN. Vương Vũ Hoàng**
- **ThS. Thái Mỹ Ngân**
- **ThS. Lê Trí Viễn**



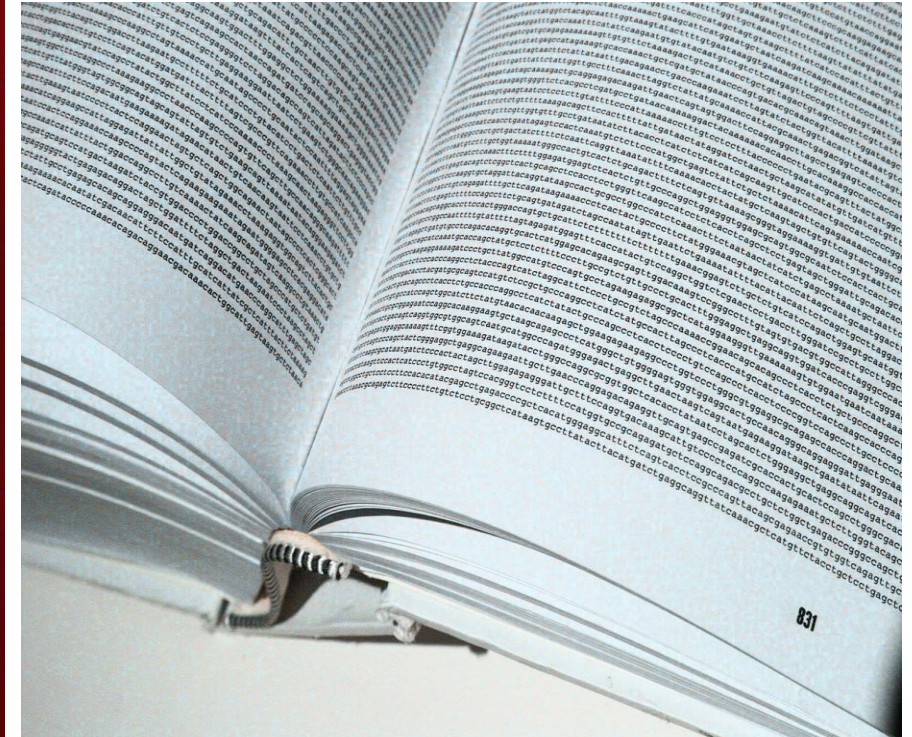
Giới thiệu

DỰ ÁN BỘ GEN NGƯỜI
& GIẢI TRÌNH TỰ GEN THỂ
HỆ MỚI

Dự án hệ gen người (1990-2003)



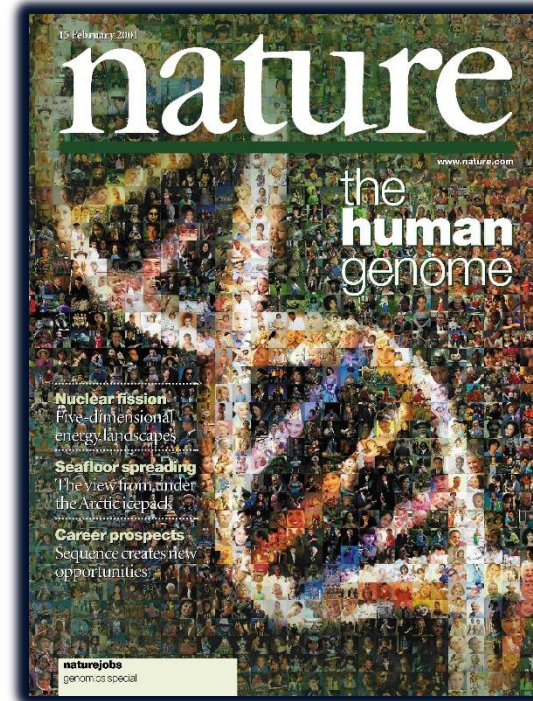
Năm 2000



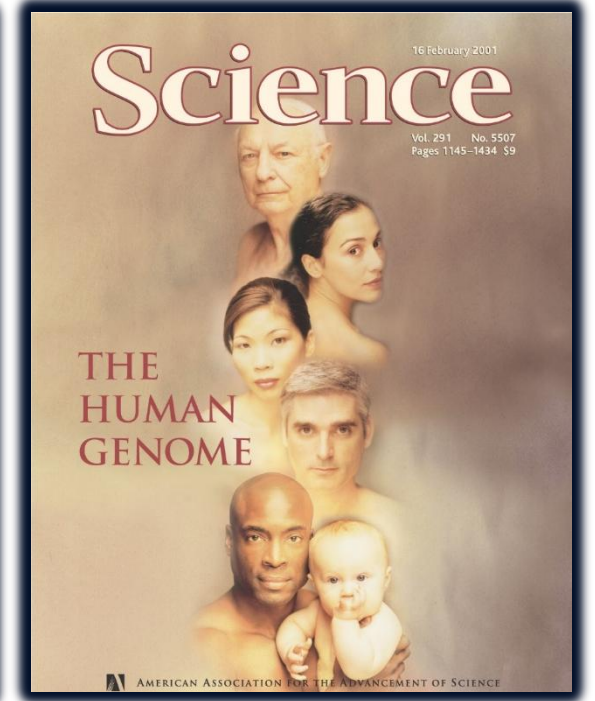
Dự án hệ gen người HGP (Oct 1990 - April 2003)

- Năm 2003, dự án hệ gen người HGP đã tạo ra một chuỗi trình tự gần 3 tỉ nucleotide chiếm hơn 90% bộ gen người.
- Đây là chuỗi gen hoàn chỉnh nhất có thể đạt được với công nghệ giải trình tự DNA thời điểm đầu những năm 2000.

=> Sự phát triển của công nghệ giải trình tự thế hệ mới (NGS).

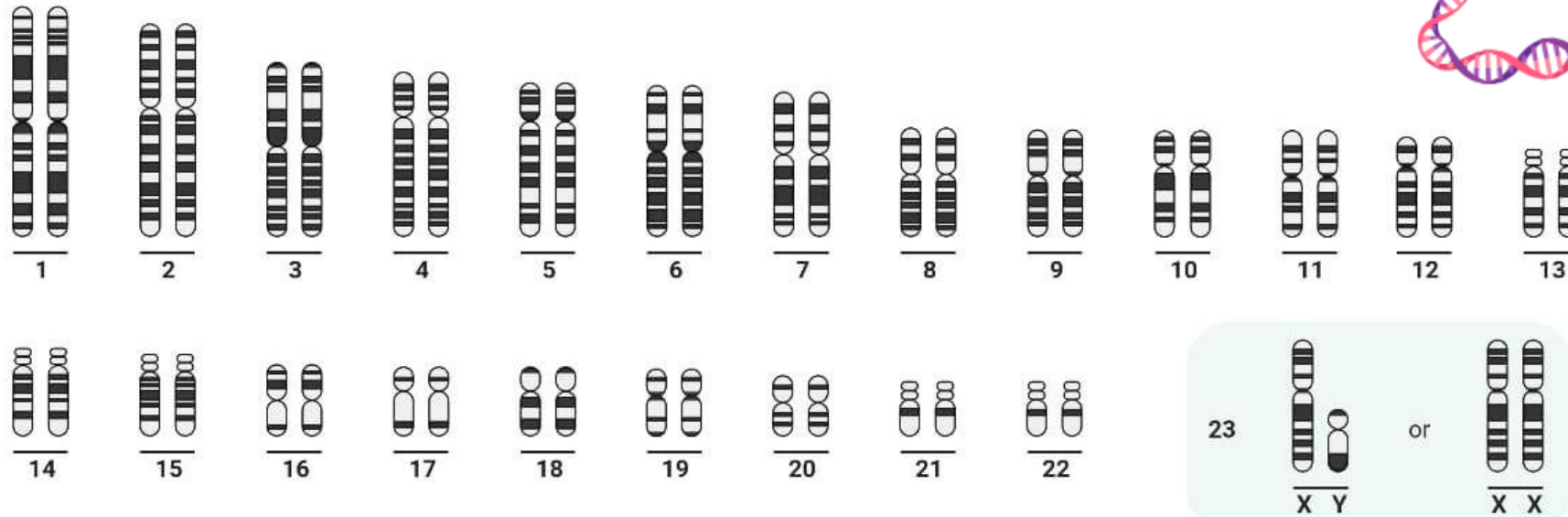
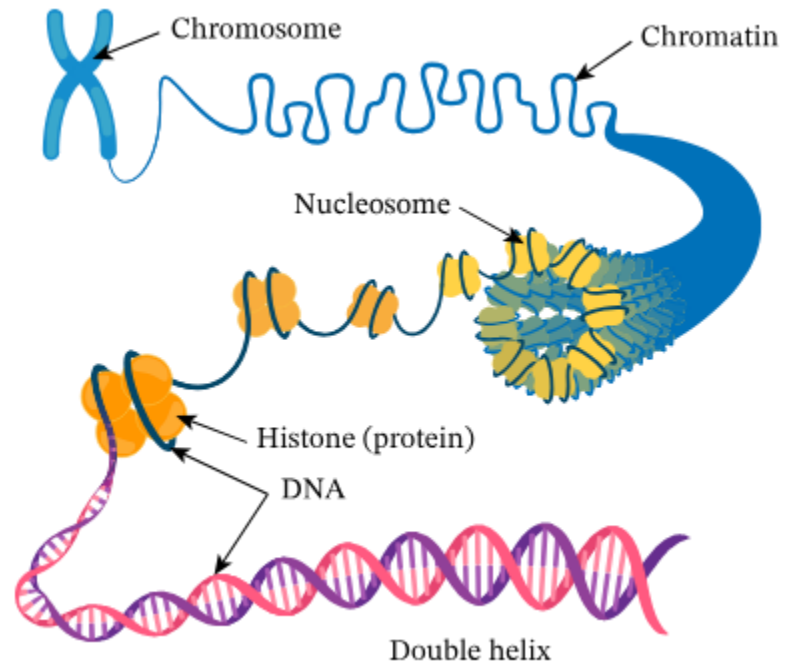


HGP Paper



Venter/Celera Paper

Human Karyotype



<https://microbenotes.com/karyotype-karyotyping/>

Hệ Gen người

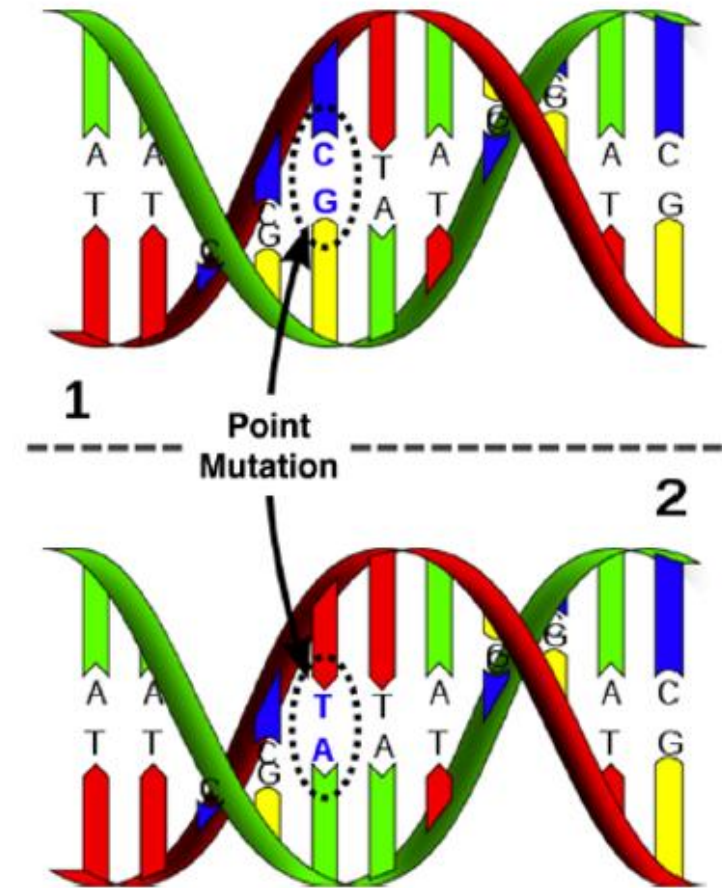
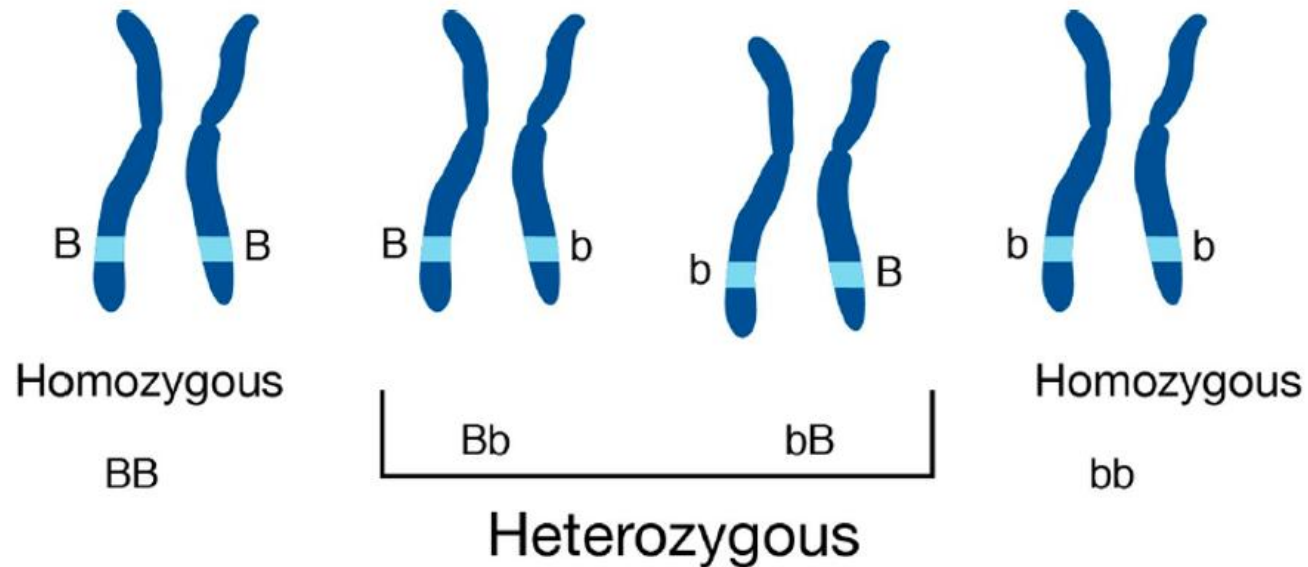


48541	agcccttcaa	agaaatgttc	tcagcaggca	tggagcccag	gacttgctcc	ctttgggtag
48601	agagccgggt	tgaagggtgac	tgaagtga	tgggacagta	gaggcggggg	gggtggtgag
48661	ttcctggagg	tgggggggtg	gggaacctgc	tgtgtactga	gatgcacccc	tgccagttct
48721	gcctgaagat	ttgaggcggg	gggcaggggg	gcggagtgaa	gtcattttac	tggtaaagtaa
48781	ttttaaacct	tttaatat	aagcaaacgt	ggatatgtaa	tgaatgaaat	tcattctgga
48841	atgaaaaatt	cacgtgatgt	tgaaaaataa	cacggggcct	cagagaggac	tttctggctg
48901	gcagcagact	ccagattccc	agggcccctg	caccctcctc	tgcccacagg	gcaccttaat
48961	ggagaagggtg	tgggaggaga	gccaggcccg	agtcagagca	cactggtgac	tccacatttg
49021	cagcgtgccc	tgctctctc	ctgaggcttg	gcaacgtgca	atatgtctag	caaactcccc
49081	ctgtccccgt	ccagtttctg	aggacaagag	ccaccacctg	tagcaataa	agaccacagca
49141	accctttgac	tcatctttgt	gagtctctg	aatcagaggg	tagccacatc	gctgagagggt
49201	ggagtgaagc	actcgggtga	aaaggtacaa	ggaagtcagg	gacaggagtg	tggggacatc
49261	acctagacaa	tgacagagaa	gaggggcaca	gccgagtgag	gggagagggg	ccggcagttc
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49741	ggcacttggc	aaacagcaaa	agagtccaaa	tgtgcaggct	gggccaggcg	cagtggctca
49801	tgcctgtaat	cccagcaatt	taggaagcca	agggtggcgg	atcacctgag	gtcaggagtt
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50401	cagccagcat	ggcccttcct	gggagcccat	ggtgggggat	tcctgctgca	gccaaggctc
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50641	acagggtgtga	cagggatggg	gactgggaag	taagagatga	aaccctggct	ggaggctgtg
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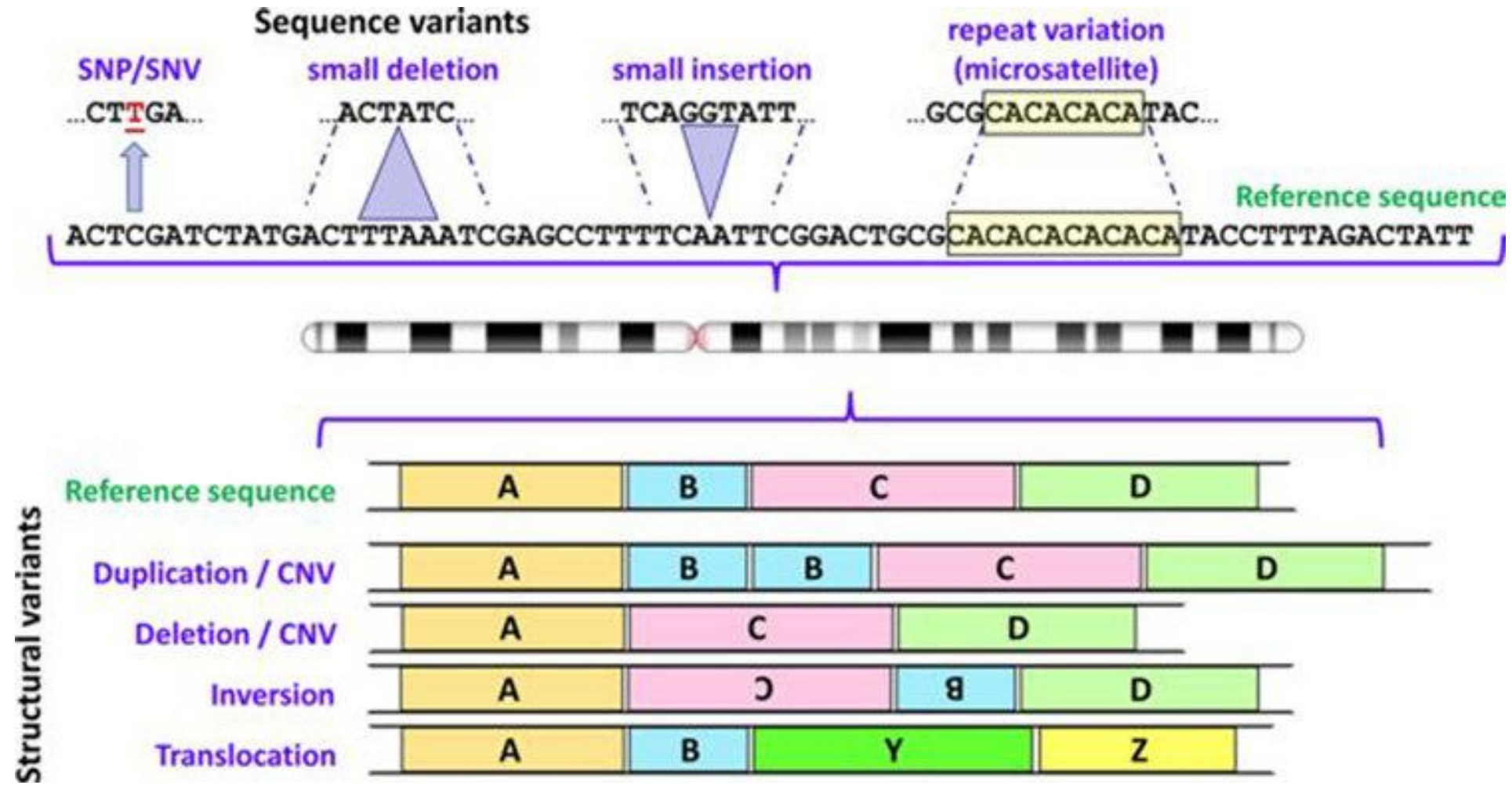
<https://www.ncbi.nlm.nih.gov/nuccore/806904736>

Các loại biến thể trên hệ gen

- Hệ gen giữa hai người giống nhau > 99%
- Mỗi người có khoảng 5 triệu biến thể, trong đó có 3 đến 4 triệu biến thể một nucleotide
- Hệ gen người là hệ lưỡng bội



Các loại biến thể trên hệ gen

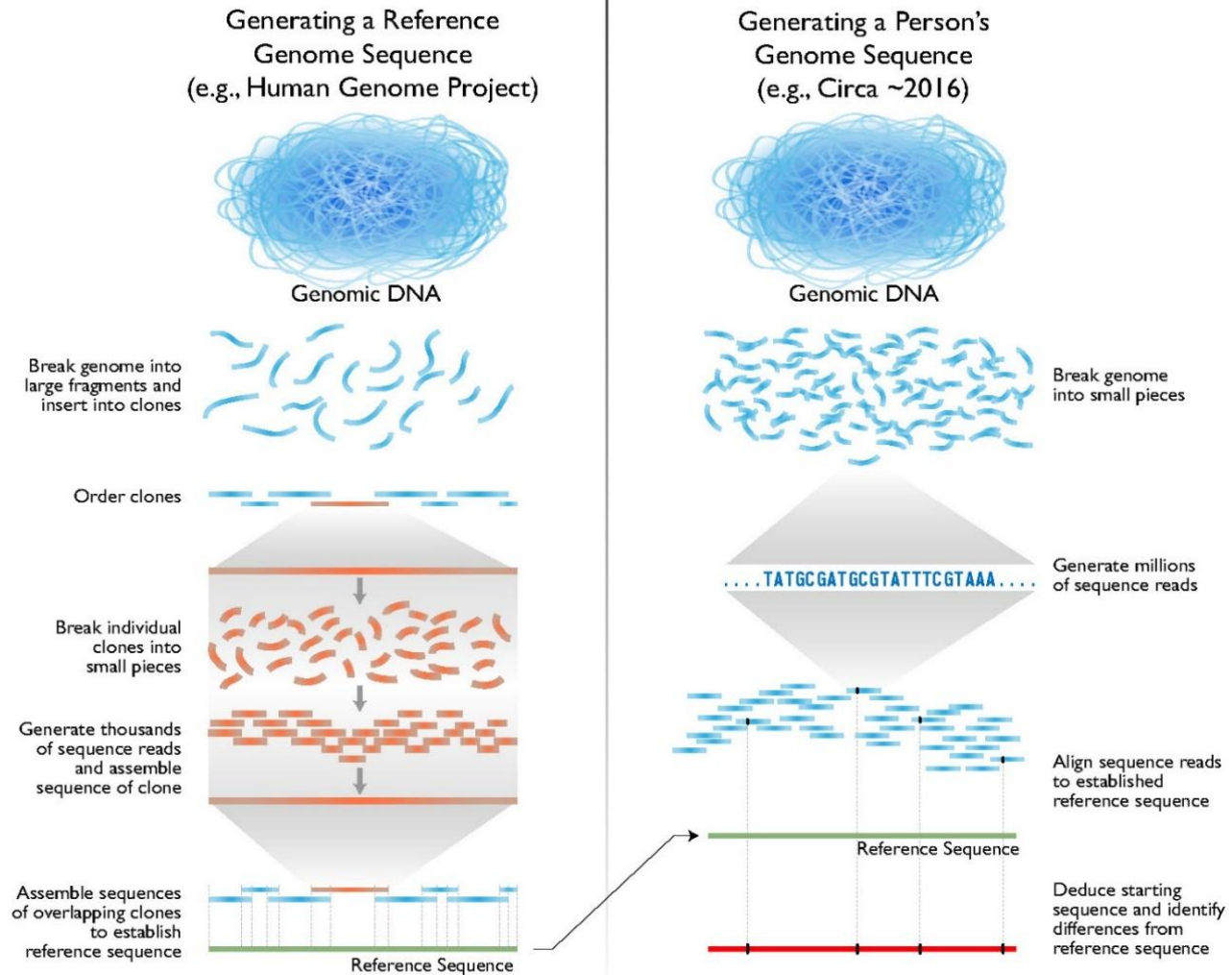


First Generation Sequencing (Sanger Sequencing)

- <https://www.youtube.com/watch?v=KTstRrDTmWI>
- <https://www.youtube.com/watch?v=X9566yI2cBo>

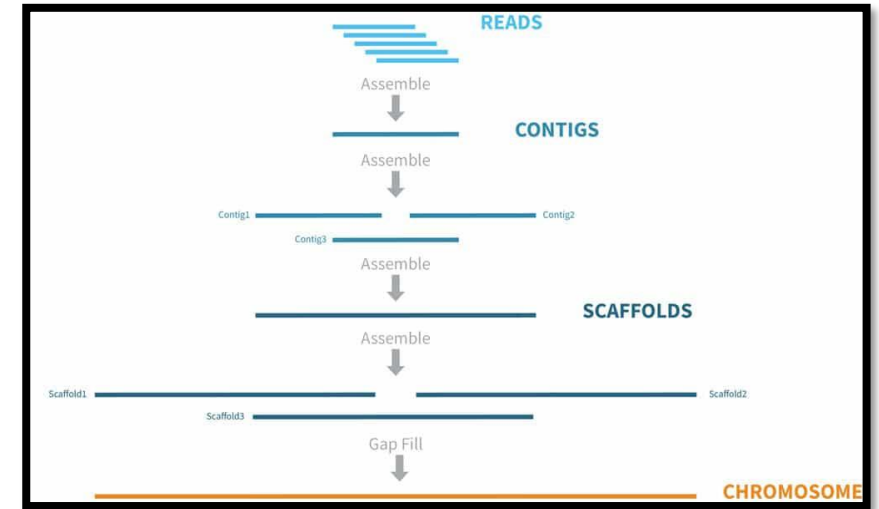
Giải trình tự gen thế hệ mới (NGS)

Human Genome Sequencing

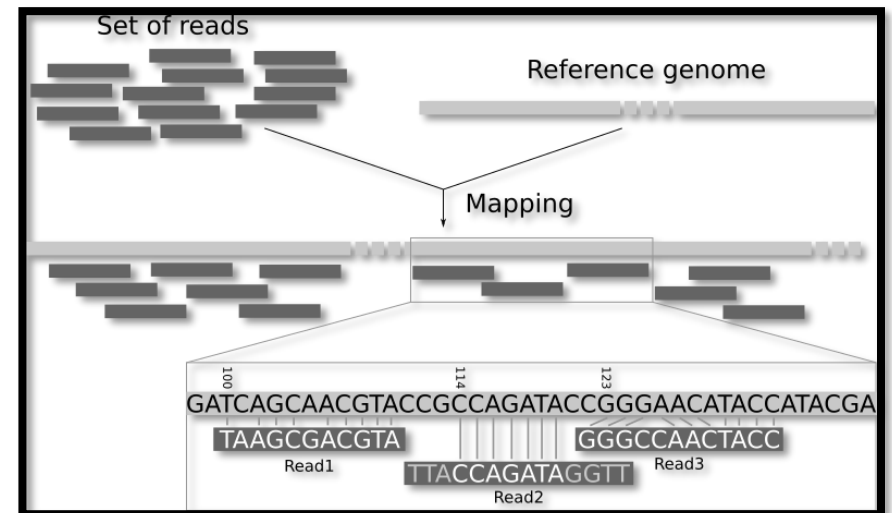


<https://www.genome.gov/about-genomics/fact-sheets/Sequencing-Human-Genome-cost>

De novo assembly



Mapping to reference



Giải trình tự gen thế hệ mới (NGS): Có hệ gen tham chiếu

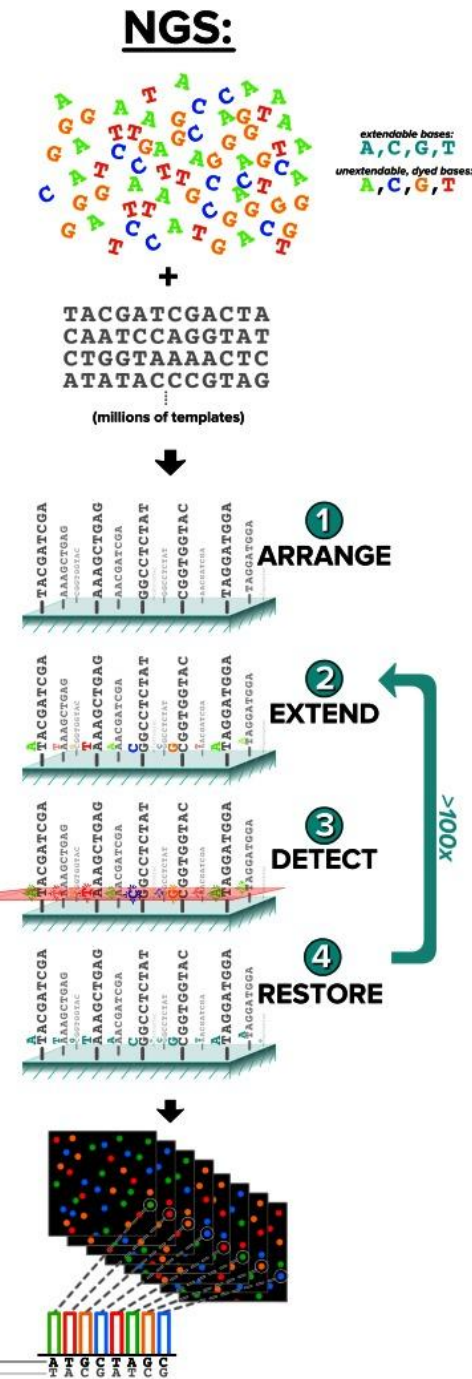
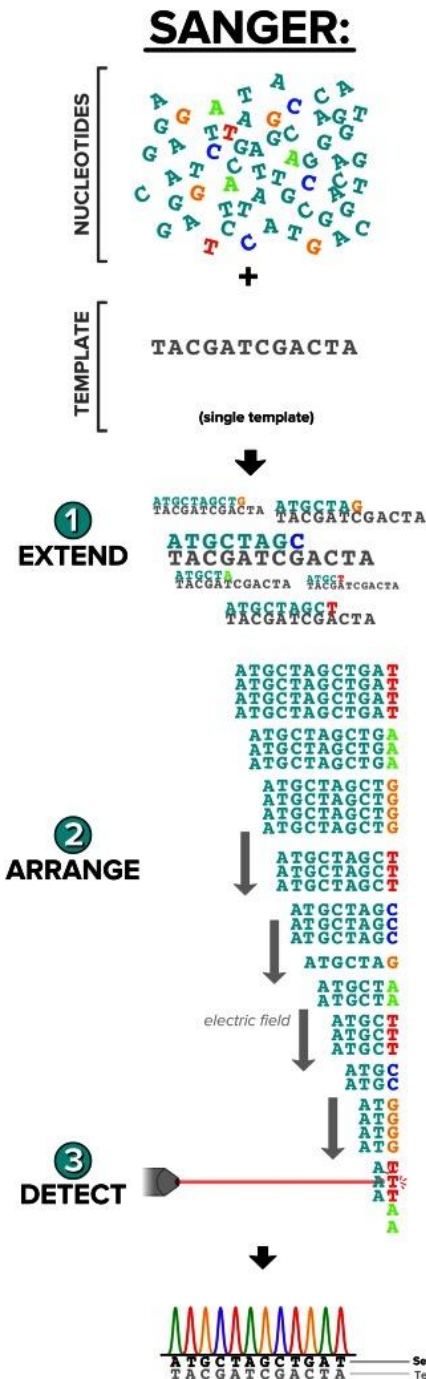
De novo assembly



Mapping to reference



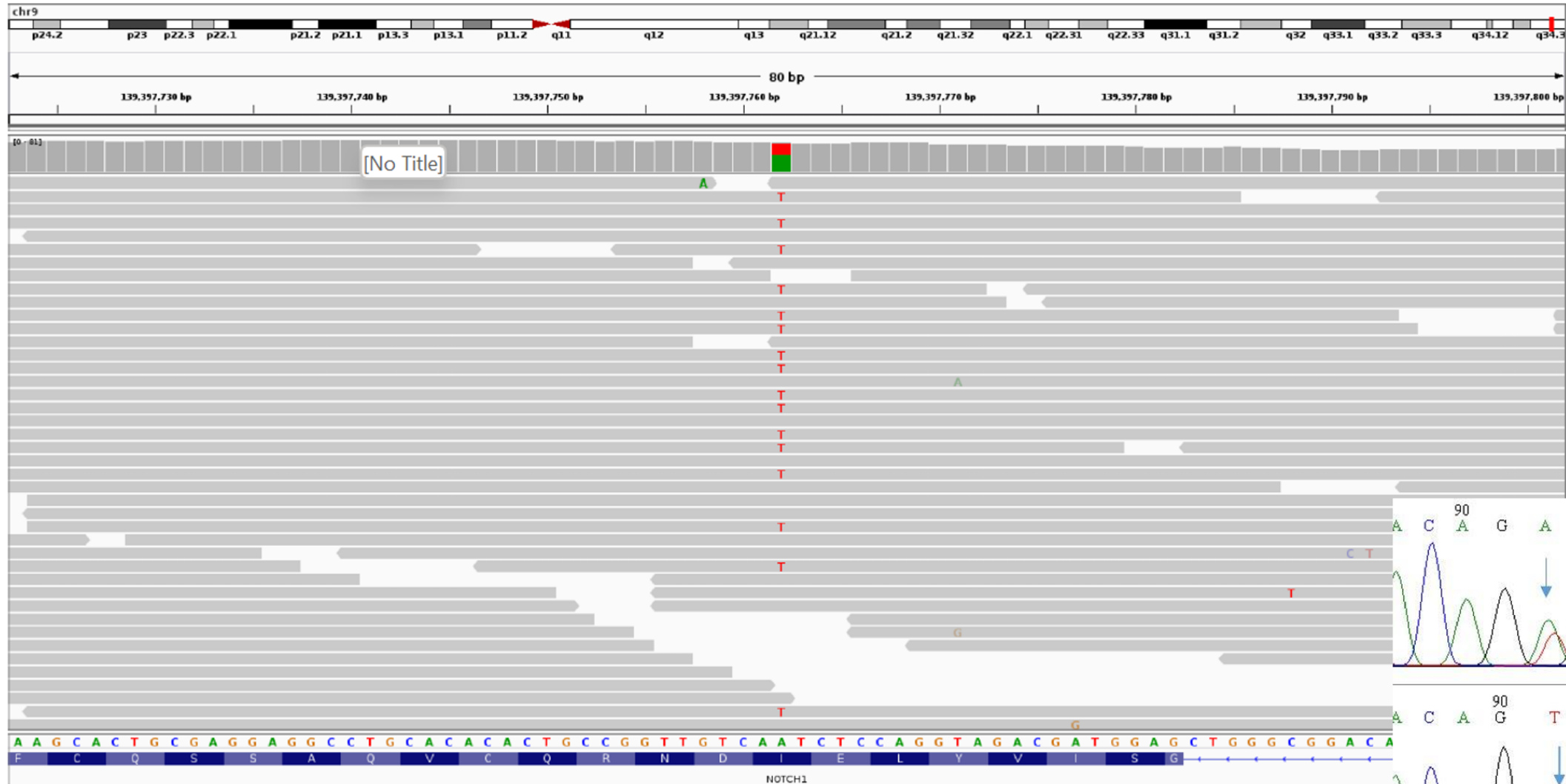
Giải trình tự gen thế hệ mới (NGS): giải trình tự song song



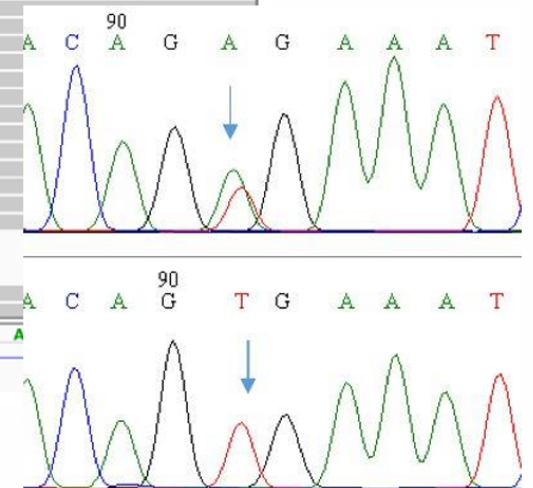
Giải trình tự gen thế hệ mới (NGS): chuẩn bị thư viện có kit thương mại

Read1: CTCGAATACG
Read2: CTCGAATACG
Read3: CTCGAATACG
Read4: CTCGAATACG
Read5: CGCGAATACG
Read6: CGCGAATACG
Read7: CGCGACTACG
Read8: CGCGAATACG

Trực quan hóa dữ liệu giải trình tự DNA bằng NGS



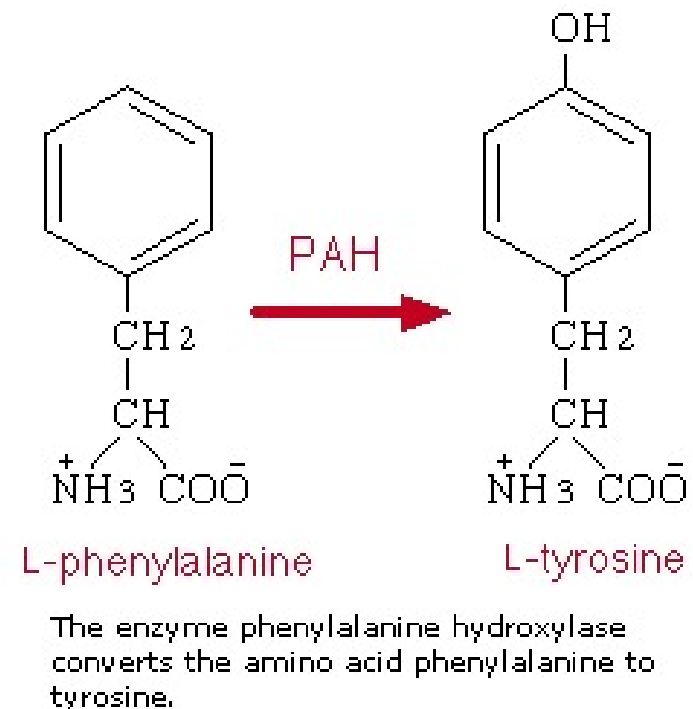
Sanger



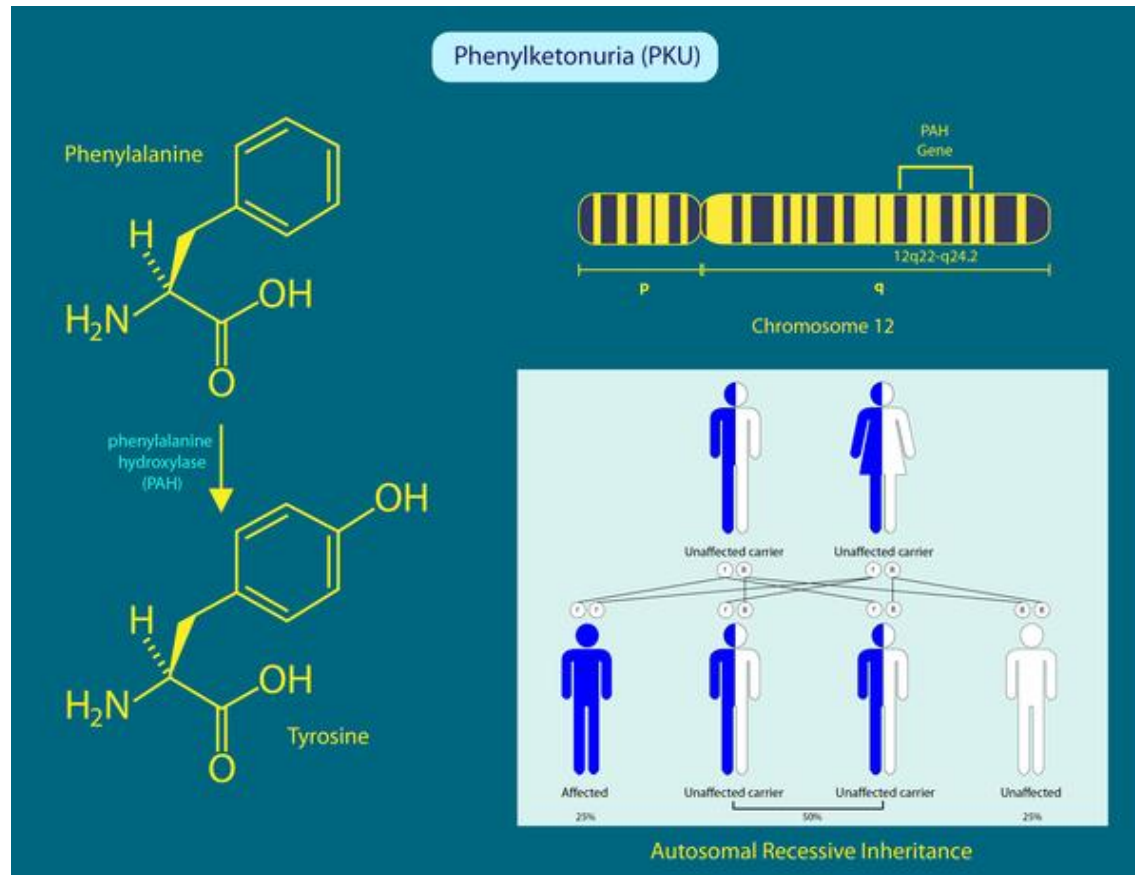
Ví dụ mối quan hệ của BIẾN THỂ GEN và BỆNH DI TRUYỀN

Phenylketonuria (PKU): Rối loạn chuyển hóa di truyền

- Nguyên nhân do thiếu hụt enzyme phenylalanine hydroxylase.
- Sự mất enzyme này dẫn đến suy giảm trí tuệ, tổn thương cơ quan, tư thế bất thường.
- Tần suất xảy ra PKU khác nhau giữa các nhóm dân tộc và các vùng địa lý trên toàn thế giới. Ở Hoa Kỳ, PKU xảy ra ở 1 trong 25.000 trẻ sơ sinh.
- Hầu hết các trường hợp PKU được phát hiện ngay sau sinh bằng sàng lọc sơ sinh và điều trị được bắt đầu ngay lập tức.



Phenylketonuria (PKU): Rối loạn chuyển hóa di truyền



Tên khác của PKU

- Folling disease
- Folling's disease
- PAH deficiency
- Phenylalanine hydroxylase deficiency
- Phenylalanine hydroxylase deficiency disease

<https://medlineplus.gov/genetics/condition/phenylketonuria/>

Trình tự gen PAH ở người - Homo sapiens (5053)

Gene (Nucleotide)

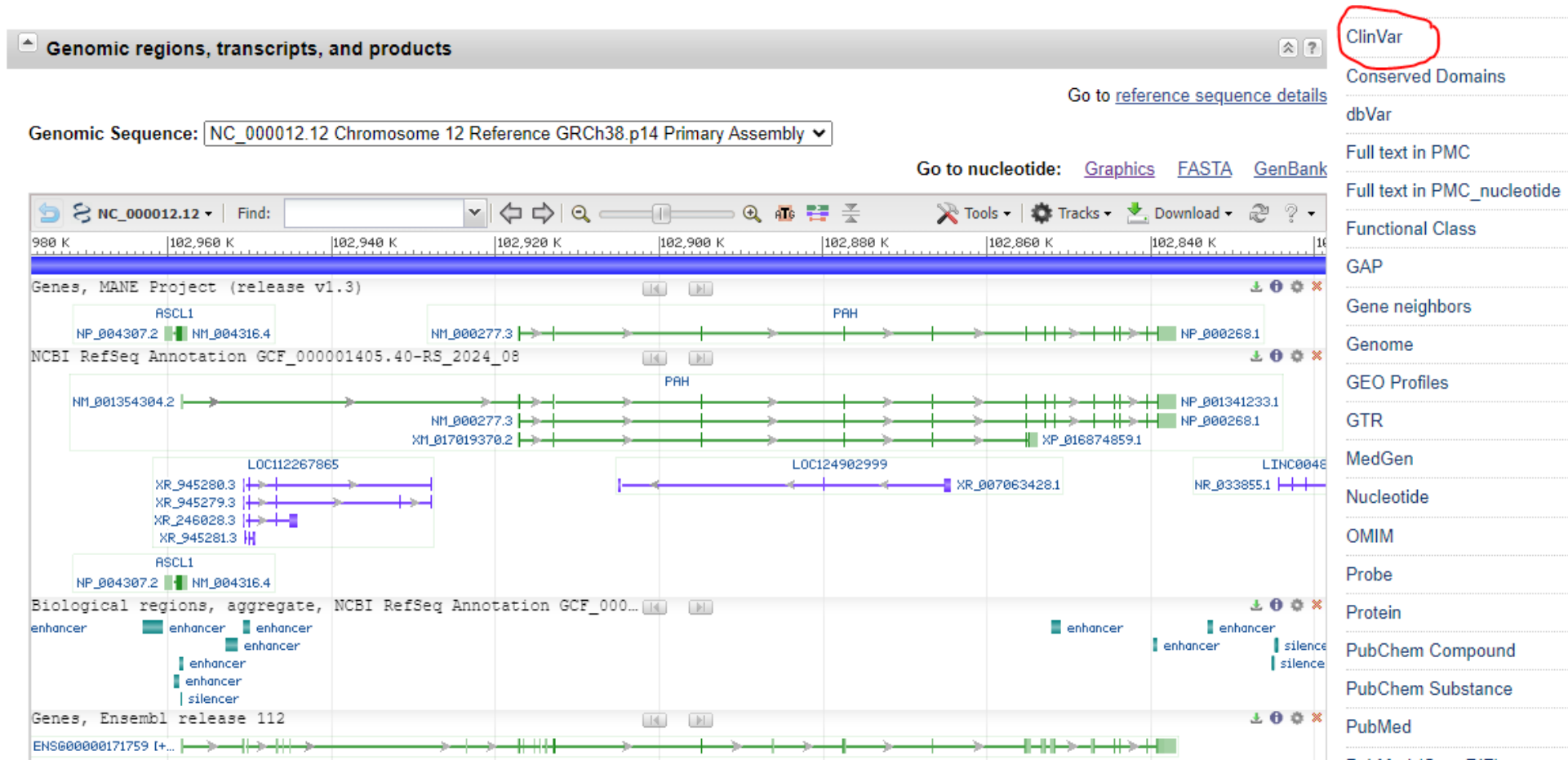
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Protein (Amino Acid)

AA seq	452 aa AA seq DB search
	MSTAVLENPGLGRKLSDFGQETSYIEDNCNQNGAISLIFSLKEEVGALAKVLRLFEENDV NLTHIESRPSRLKKDEYEFFTHLDKRSLPALTNIKILRHDIGATVHELSDRKKKDTV FPRTIQELDRFANQILSYGAELDADHPGFKDPVYRARRKQFADIAYNYRHGQPIPRVEYM EEEKKTWGTVFKTLKSLYKTHACYEYNHIFPILLEKYCGFHEDNIPQLEDVSQFLQCTGF RLRPVAGLLSSRDFLGGLAFRVFHCTQYIRHGSKPMYTPEDICHELLGHVPLFSDRSFA QFSQEIGLASLGAPDEYIEKLATIWFTVEFGLCKQGDSIKAYGAGLLSSFGE LQYCLSEKPKLLPLELEKTAIQNYTVTEFQPLYYVAESFNDAKEKVRNFAATIPRPF SVRYDPYTQRIEVLNTQQLKILADSINSEIGILCSALQKIK

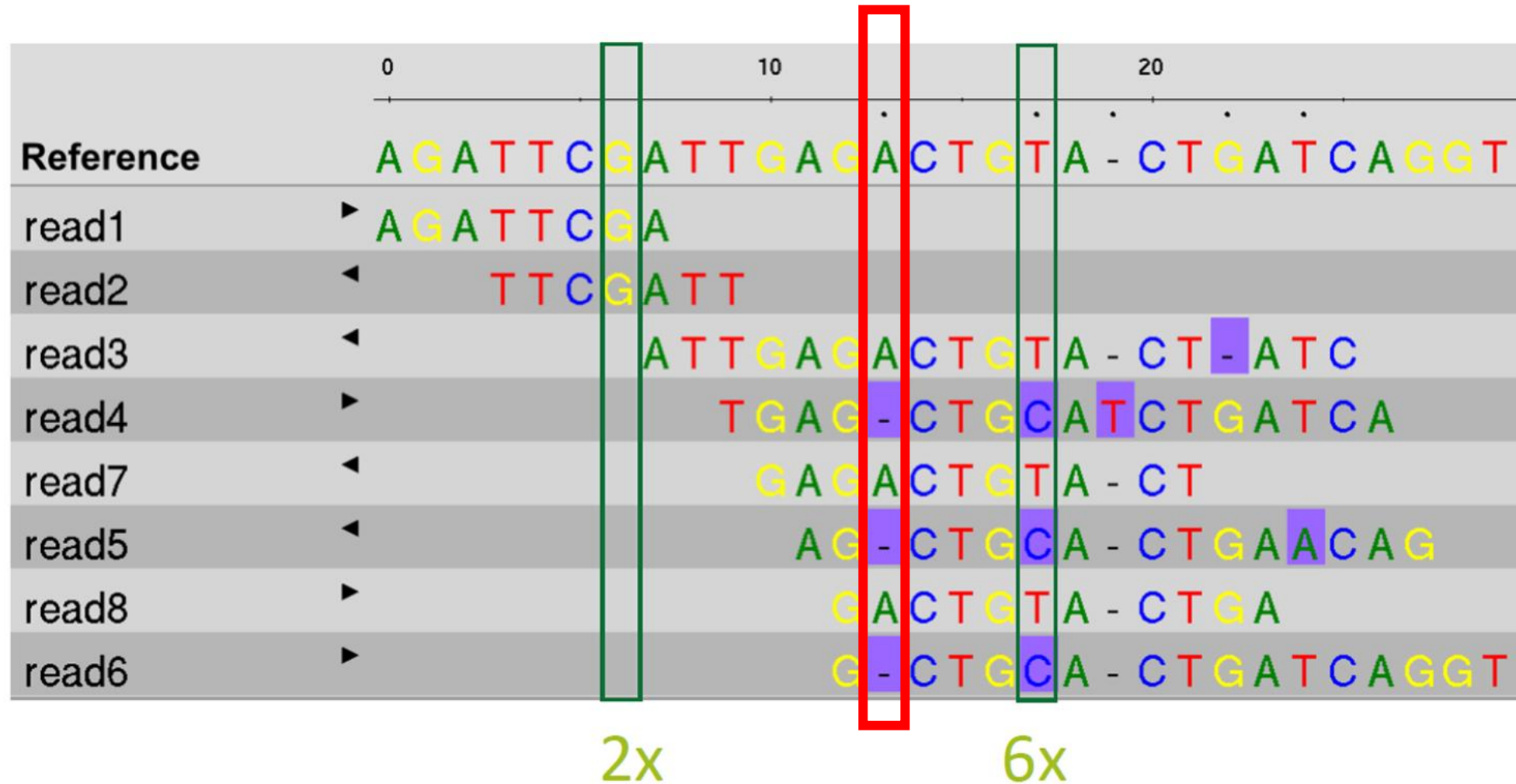
<https://www.genome.jp/entry/T01001:5053>

Gen PAH (Năm 2009)



<https://www.ncbi.nlm.nih.gov/gene/5053>

Variants in PAH



Cơ sở dữ liệu ClinVar (Năm 2013) cho gen PAH

Classification type

- ☐ Germline (208)
- ☐ Somatic (0)

Germline classification

- ☐ Conflicting classifications (2)
- ☐ Benign (10)
- ☐ Likely benign (21)
- ☐ Uncertain significance (41)
- ☐ Likely pathogenic (40)
- ☐ Pathogenic (39)

Types of conflicts

- ☐ P/LP vs LB/B (0)
- ☐ P/LP vs VUS (0)
- ☐ VUS vs LB/B (2)

Molecular consequence

- ☐ Frameshift (20)
- ☐ Missense (96)
- ☐ Nonsense (7)
- ☐ Splice site (9)
- ☐ ncRNA (0)
- ☐ Near gene (0)
- ☐ UTR (27)

Variation type

- ☐ Deletion (39)

Links from Gene

[Display options](#) [Sort by Relevance](#) [Download](#)

Items: 1 to 100 of 209

<< First < Prev Page 1 of 3 Next > Last >>

Variation	Gene (Protein Change)	Type (Consequence)	Condition	Classification, Review status
☐ NM_004316.4(ASCL1):c.51G>T.(p.Gln17His)	ASCL1, PAH (Q17H)	Single nucleotide variant (missense variant +1 more)	not specified	G Uncertain significance ★
☐ NC_000012.11:g.(?_103232953)_ (1_03240749_?)del	PAH	Deletion	Phenylketonuria	G Pathogenic ★
☐ NC_000012.11:g.(?_103288493)_ (1_03310908_?)del	PAH	Deletion	Phenylketonuria	G Pathogenic ★
☐ NC_000012.11:g.(?_103248894)_ (1_03249131_?)del	PAH	Deletion	Phenylketonuria	G Pathogenic ★
☐ NC_000012.12:g.(?_102894715)_ (102894938_?)del	PAH	Deletion	Phenylketonuria	G Pathogenic ★
☐ NC_000012.11:g.(?_103306549)_ (1_03306696_?)del	PAH	Deletion	Phenylketonuria	G Pathogenic ★
☐ NM_000277.3(PAH):c.1179_1180del (p.Asn393fs)	PAH (N393fs)	Deletion (frameshift variant)	Phenylketonuria	G Likely pathogenic ★

https://www.ncbi.nlm.nih.gov/clinvar?LinkName=gene_clinvar&from_uid=5053

Biến thể gây bệnh - Pathogenic variant in PAH

NM_000277.3(PAH):c.971T>A (p.Ile324Asn)

ClinVar Genomic variation as it relates to human health

Search by gene symbols, location, HGVS expressions, c-dot, p-dot, conditions, ; **Search ClinVar** ?

Advanced search

About Access Submit Stats FTP Help

NM_000277.3(PAH):c.971T>A (p.Ile324Asn) Cite Follow Print Download

We've updated the ClinVar website to better support classifications of somatic variants!
Read more about changes to the website in our [web release notes](#); more information about somatic variants in ClinVar is available on [GitHub](#).

Germline

Top reviewed classifications are shown here. Submission summary: [1 submission](#) [1 submitter](#) [1 condition](#)

Reviewed by expert panel
★★★★☆

Pathogenic
Dec 2023 by ClinGen PAH Va...
FDA RECOGNIZED DATABASE

for Phenylketonuria

☒

Somatic

No data submitted for somatic clinical impact

Somatic

No data submitted for oncogenicity

☐

Feedback

On this page
[Classification Summary](#)
[Variant Details](#)
[Genes](#)
[Germline](#)
[Conditions](#)
[Submissions](#)
[Citations](#)
[Text mined Citations](#)

<https://www.ncbi.nlm.nih.gov/clinvar/variation/2682170/>

Các biến thể gây bệnh trên gen PAH

Gene: PAH

[View on UniProt](#)

[View on SwissModel](#)

Transcript: ENST00000553106.6

Select protein structure

SwissModel:5den 20-450 (number o... ▼

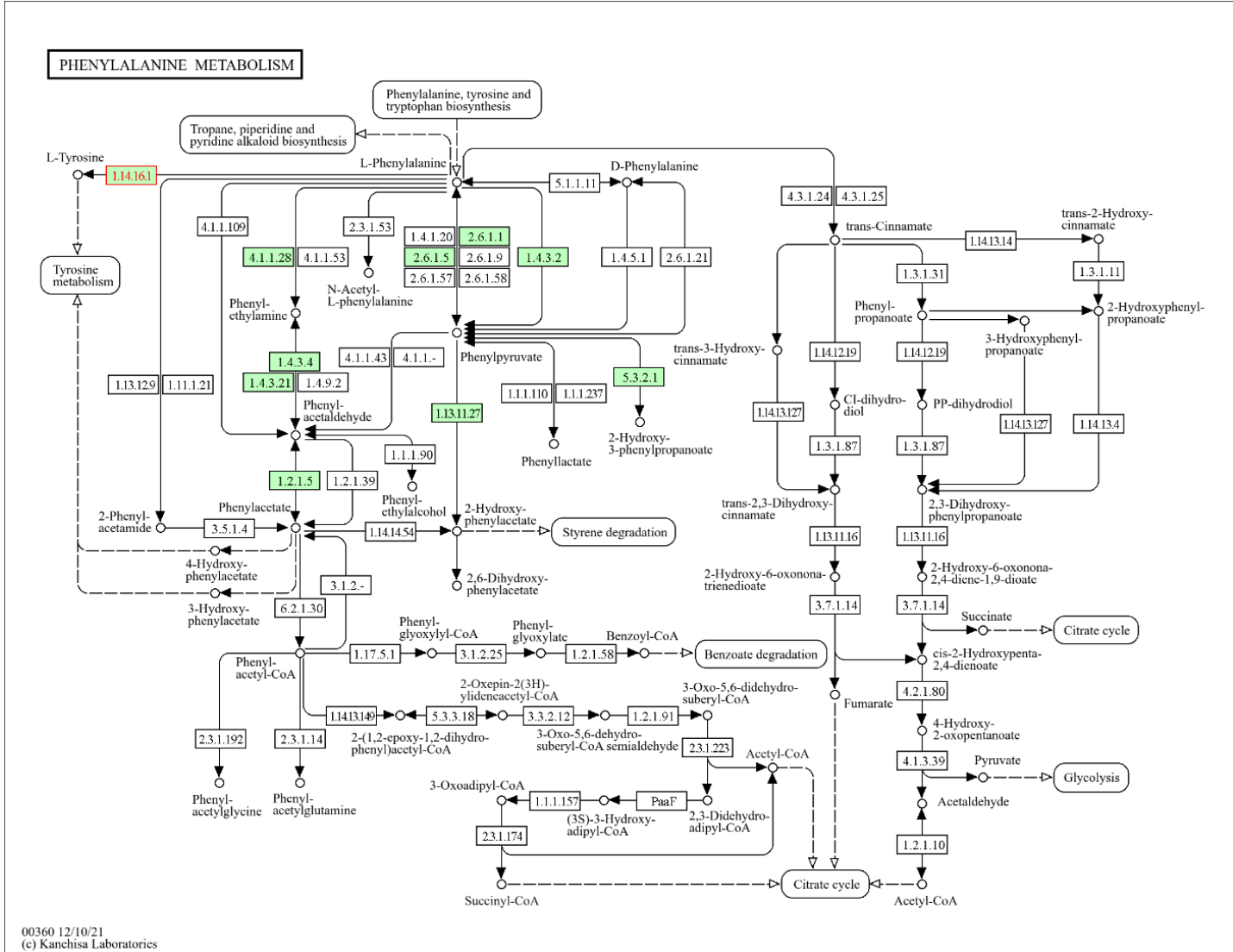


Show

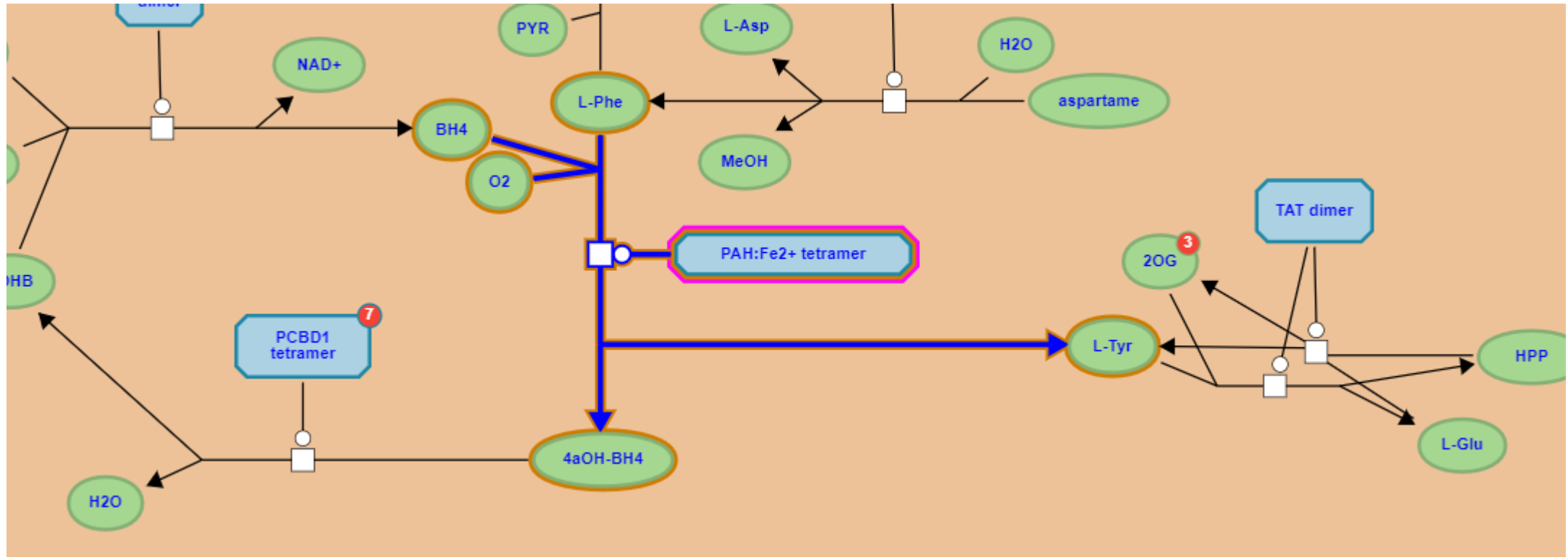
- ☐ All Residues
- ☒ Variants
 - ☒ Pathogenic
 - ☐ Likely Pathogenic
 - ☐ Uncertain Significance
 - ☐ Likely Benign
 - ☐ Benign
 - ☒ Current Variant

<https://varsome.com/variant/hg38/chr12%3A102844430%3AA%3AT?>

PAH: chuyên hóa Phenylalanine thành Tyrosine

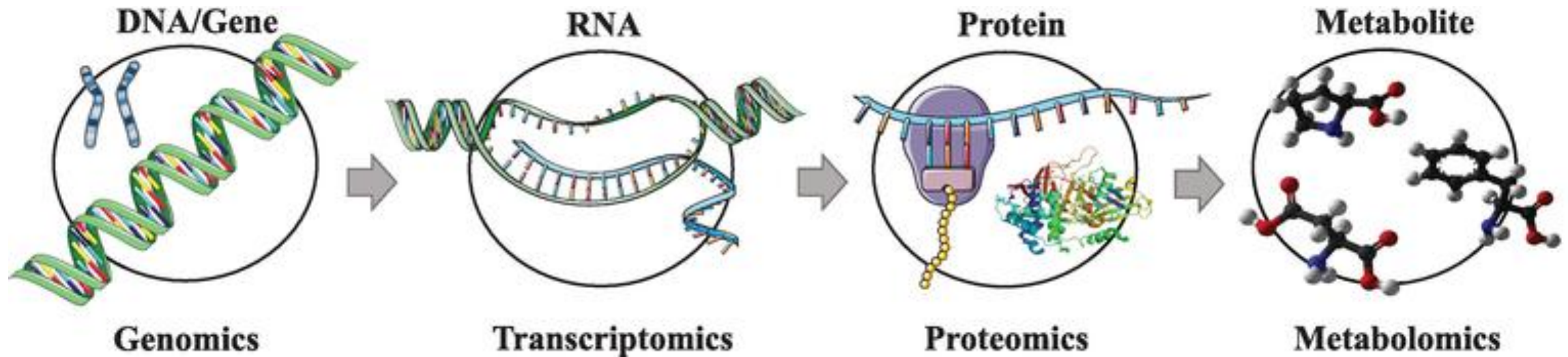


PAH: chuyển hóa Phenylalanine thành Tyrosine



<https://reactome.org/PathwayBrowser/#/R-HSA-8963691&SEL=R-HSA-71118&PATH=R-HSA-1430728,R-HSA-71291&FLG=UniProt:P00439>

Mối liên kết: Biểu thể gen và bệnh di truyền



PAH gene

Ref ...A**T**CGAT...
P1 ...A**A**CGAT...

NM_000277.3(PAH):c.971T>A

PAH mRNA

Ref ...A**U**CGAU...
P1 ...A**A**CGAU...

NM_000277.3(PAH):c.971T>A

PAH protein

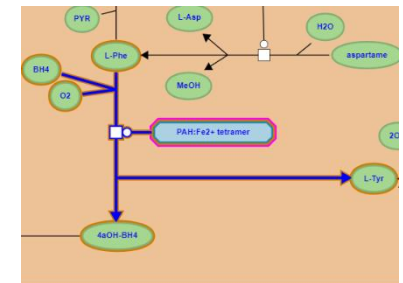
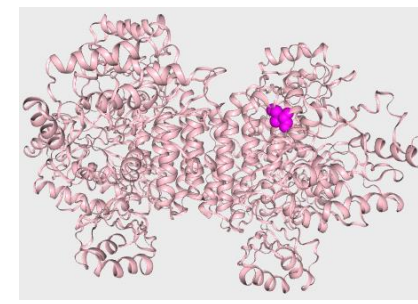
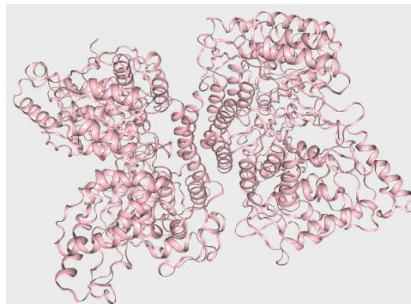
Ref ...**Ile**-Asp...
P1 ...**Asn**-Asp...

NM_000277.3(PAH):p.Ile324Asn

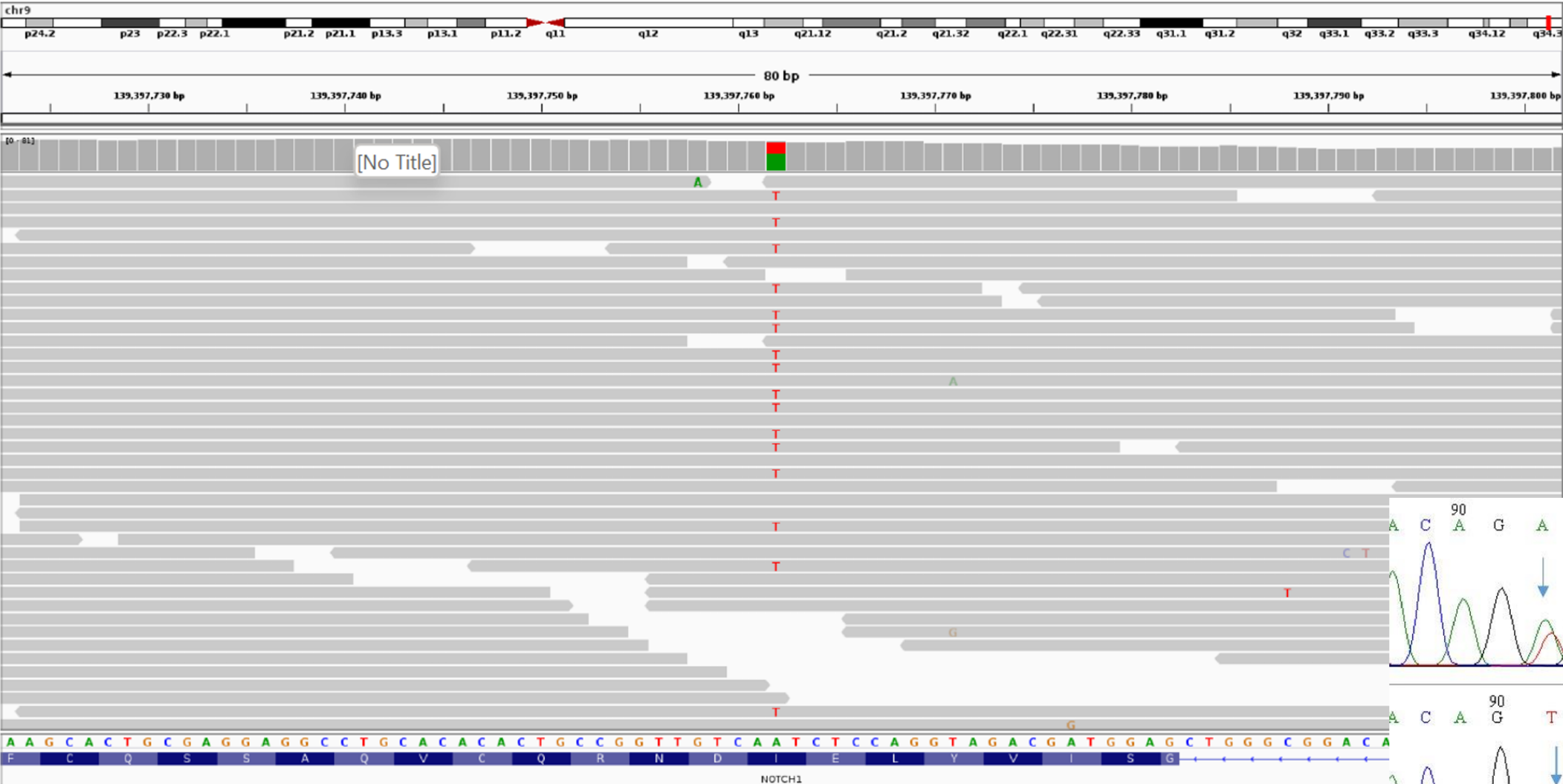
PAH

Ref Phe → Tyr

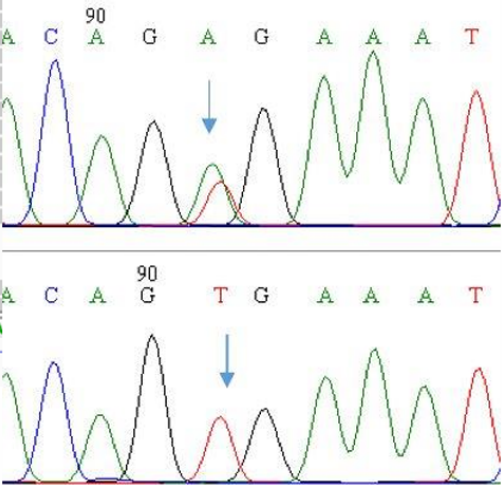
PAH
P1 Phe ~~→~~ Tyr



Trực quan hóa dữ liệu giải trình tự DNA bằng NGS



Sanger



Dữ liệu hệ gen

A G T A T A A G G T G C A T T C C C
 A C A A A A A G T A T A A G G T G C A T T C
 C C A T A A G T C C G A T A A G T C C C C T C G C A T T C C G G T
 A A A G T A T A A G G T G C A T T C C G A A G G G A T A A A G T C
 G A T C C C A T A A G G G T G C A T T C C C A G A C
 A C A A A T C C C A T C C C A T C C C A T C C G A T G



A G A G T A A G G T G A C T T C C G C A T A C G A G T C
A T C A A C T A T A G G T C A T T C C A G A T G C C A T A A A T T C G
G A T C C C A T A A G G G T G C A T T C C A G A T C G
C A C A A G T C A G G G A T A T T A G G G T C G C A T T C C C G G G T
A A A C T A A G G C G A T T C C C A G A T T C C C A T A

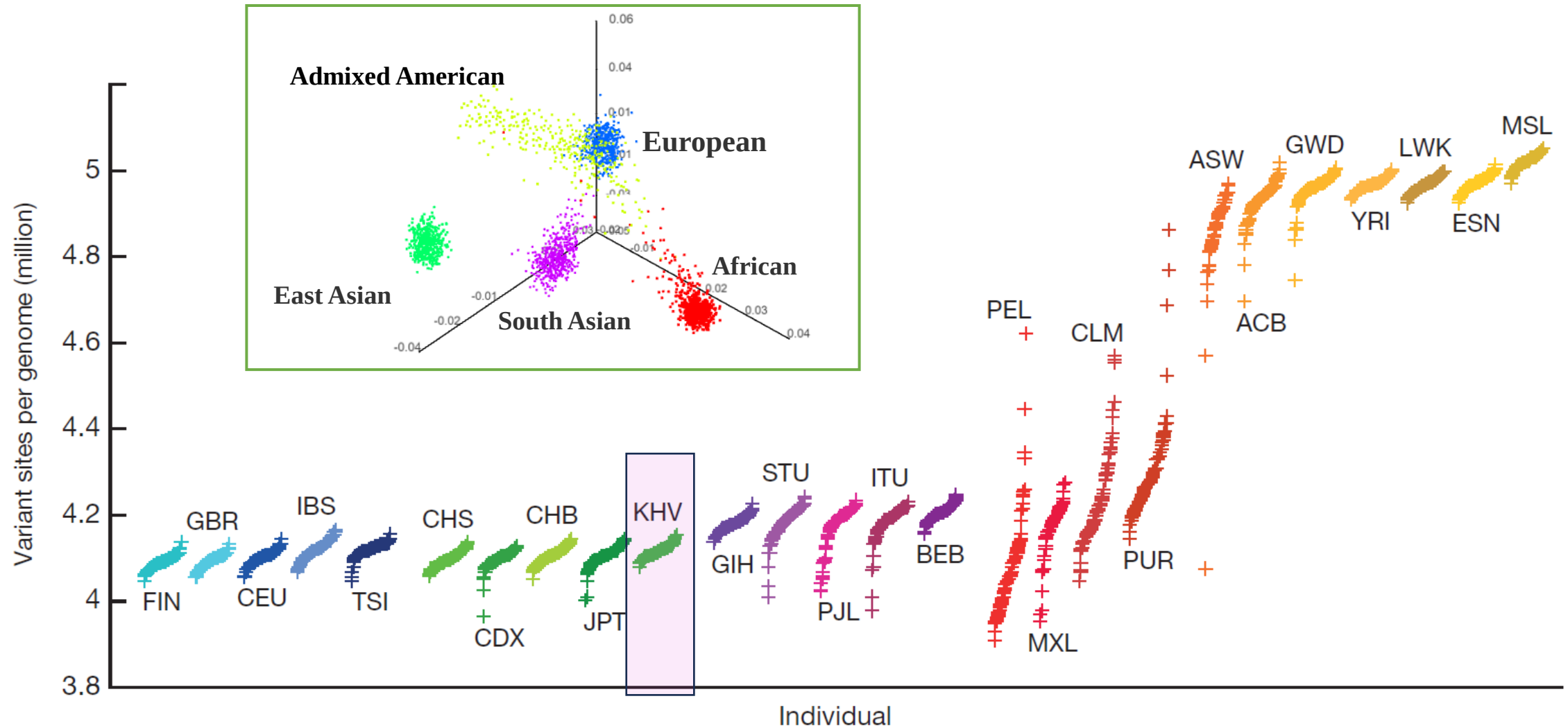


G A T A A A T C G A A T A T A G G A G T E G G G A T T C C A G A T A T C
 A C T T A A C C C A G T A T A A G G T C C A T T C C C A G A A G T C C C C A T A
 A A G T C G A A T A A G G G T G C A T T C C C A G A T C
 C A T T T C C A G A T T C C C G G A T T A G T E T E T T C E E T C C
 A C T T C C A A G T C A G G A T T A G T E T E T T C E E T C C
 G G A T T A G T A T T A G G T G C A T T G G G G
 G C A T T C C C C C C G G G T C A C C C G G
 A C A A A G T C A G G G A T A

48541	agcccttcaa	agaaatgttc	tcagcaggca	tggagccag	gacttgctc	ctttgggtag
48601	agagccgggt	tgaaggtgac	tgaagtga	tgggacagta	gaggcgggg	gggtgggtag
48661	ttctctgagg	tgggggggtg	gggaacctgc	tgtgtactga	gatgcacctc	tgccagttct
48721	gcctgaagat	ttaggcgcgg	gggcaggcgg	cggaatgtaa	gtcattttac	tggtaatgta
48781	ttttaaacat	tttaagtata	aagcaaacgt	ggatatgtaa	tgaatgaaat	tcatcttgga
48841	atgaaaaatt	cacgtgatgt	tgaaaaataa	cacggggcct	cagagaggac	tttctgctg
48901	gcagcagact	ccagattccc	agggccctg	caccctctc	tgcccacagg	gcaccttaa
48961	ggagaaggtg	tgggaggaga	gccaggcgg	agtcagagca	cactggtgac	tccacattg
49021	cagctgcgcc	tgcctctctc	ctgaggttg	gcaactgtca	atatgtcaag	caaacctccc
49081	ctgtccccgt	ccagtttctg	aggacaagag	caccacctg	tagcaataaa	agaccagca
49141	accctttgac	tcacttttgt	gagtctctgg	aatcagaggg	tagccacatc	gctgagaggt
49201	ggagtgaagc	actcgggtga	aaaggtacaa	ggaagttag	gacaggagtg	tggggacatc
49261	acctagacaa	tgacagagaa	gaggggcaca	gccgagtagg	gggagagagc	ccggcagctc
49321	tacatccctt	ggcctgaagc	acgctccagg	gcgaaaggaa	aaacactgtg	tttggggtcc
49381	aagagacagt	agttcaaatg	ctgtctcac	catctgacc	ctgtgtaacc	tgaactgtc
49441	gctgcctgaa	cctcaggttc	cccttctaaa	aatagaggag	aaaaggatgc	atttctcct
49501	gcccctgtga	gaacgaaatg	gtgcaagcac	caaggagcct	cagcaaaggt	cgggctgtcc
49561	cccgcctggc	caaaccttct	ctcttcagga	gcccacggca	accgtagttt	gacagaagag
49621	cagcaccttg	atttaatgtc	tcccagcatg	tgtccttgag	caagtcacct	aaacctctgt
49681	ggctgtcttc	tcattgggaa	aatatggctg	ccagtaaaac	ctgccctgtc	cacctctgtg
49741	ggcacttggc	aaacagcaaa	agagttcaaa	tgtgcaggct	gggcaggcgg	cagtggtctc
49801	tgcttgaat	cccagcaatt	taggaagcca	aggtggggcg	atcacctgag	gtcaggagtt
49861	tgagaccagc	ctggccaaca	tggtgaaacc	ttgtctctac	aaaaatacaa	aaatagccg
49921	ggcatgatgg	cggtgtcggt	taatcccgat	tactcgggag	gctgaggcaa	gagaatcgct
49981	tgaaccggga	aggggaagtg	tgcagtgagc	caagattgtg	ccactgcact	ccagctctgg
50041	caacagagcg	agactctgtc	tcaaaaaaaa	aaaaaaaaaa	aaacaattga	gagctggctg
50101	tgtaaaaaac	ctgttccact	gcaggggcca	gtgtccacca	ggctgggggt	caggcctatg
50161	gggtgggggc	ccagcatcag	cctctcagca	gccttgggag	gcggggcgca	tcccgtgcc
50221	ctcgtgtgtc	ggaatgtgtt	tagcccaagt	cctaggttac	acctgcgctc	gcttggcttc
50281	tcaggagagg	cccagggtga	ggaggagcat	ggtaaagggt	aaagctgatt	ggaagtcagc
50341	tgttgggaaa	gcaacctctt	gcacatttga	ggaaccgaga	aagactgacc	cgaaggacag
50401	cagccagcat	ggcccttctc	gggagcccat	tgtgggggat	tcttctgtca	gccaaagctc
50461	agcccttgtg	gtcgcagggt	ctggctctgg	cctcttcccc	tcccatgcag	gagcacagga
50521	gagatgctt	ctgaggacct	gttgacgtg	tggccttggg	aatagatttg	ccagggagct
50581	ttaaagcagc	tgagtgtgtc	atccagctaa	gctctgggga	ggagcttggc	tcaggctctg
50641	acaggtgtga	cagggatgtg	gactgggaag	taagagatga	aaccttggct	ggaggtctgt
50701	agctttcaca	gccagcgctg	gacagggagg	tgcagatat	accactagtg	gcctcaccgc

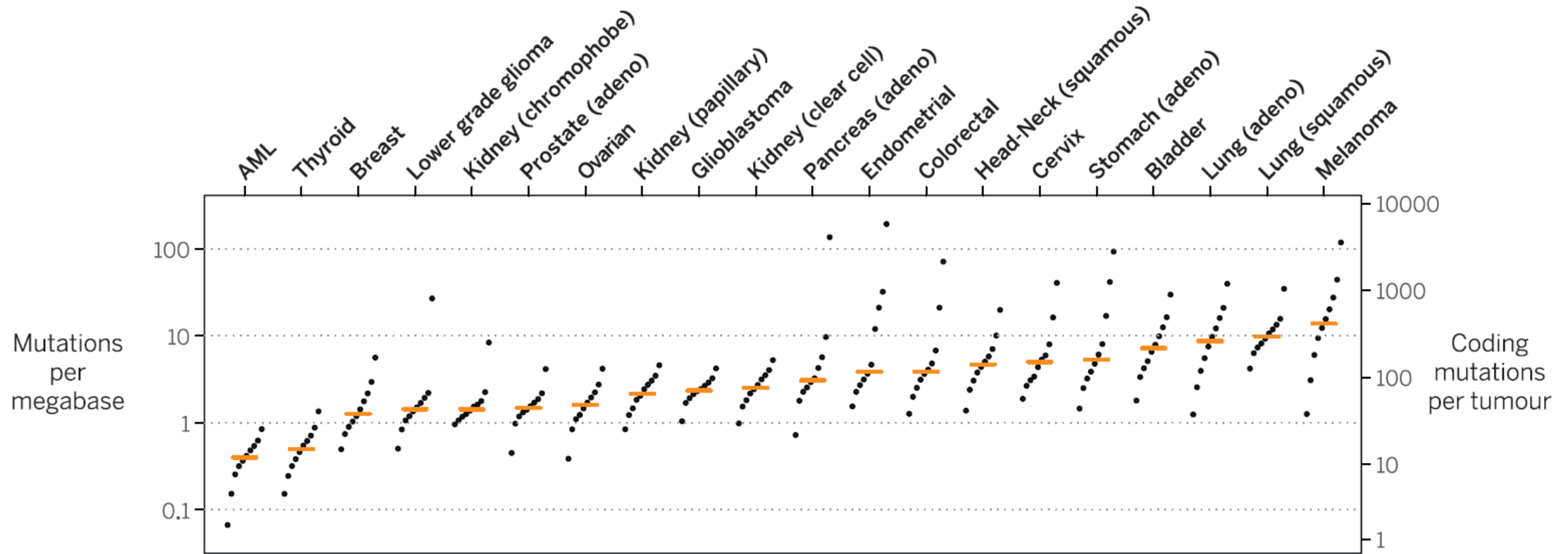


Human Genome Variation: 1000 Genomes Project



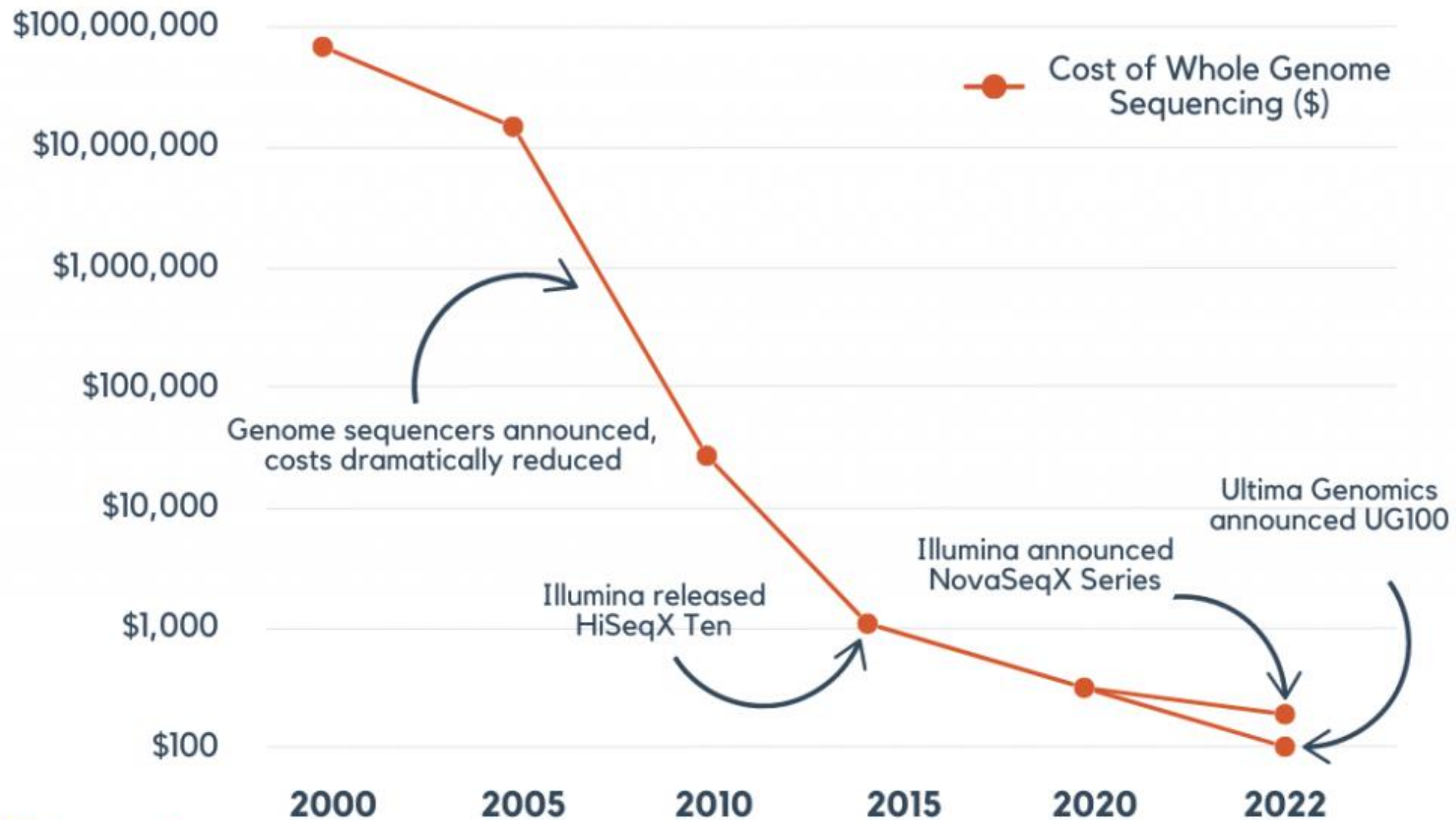
The number of variant sites per genome of 1K human genomes project (2015)
Kinh in Ho Chi Minh City, Vietnam (KHV)

Cancer Genome Somatic Variation



Mutation burden in 20 tumor types and relative contribution of different mutational processes. For each tumor type, samples were divided into deciles on the basis of their mutation burden. (2015)

Decreasing Genome Sequencing Costs

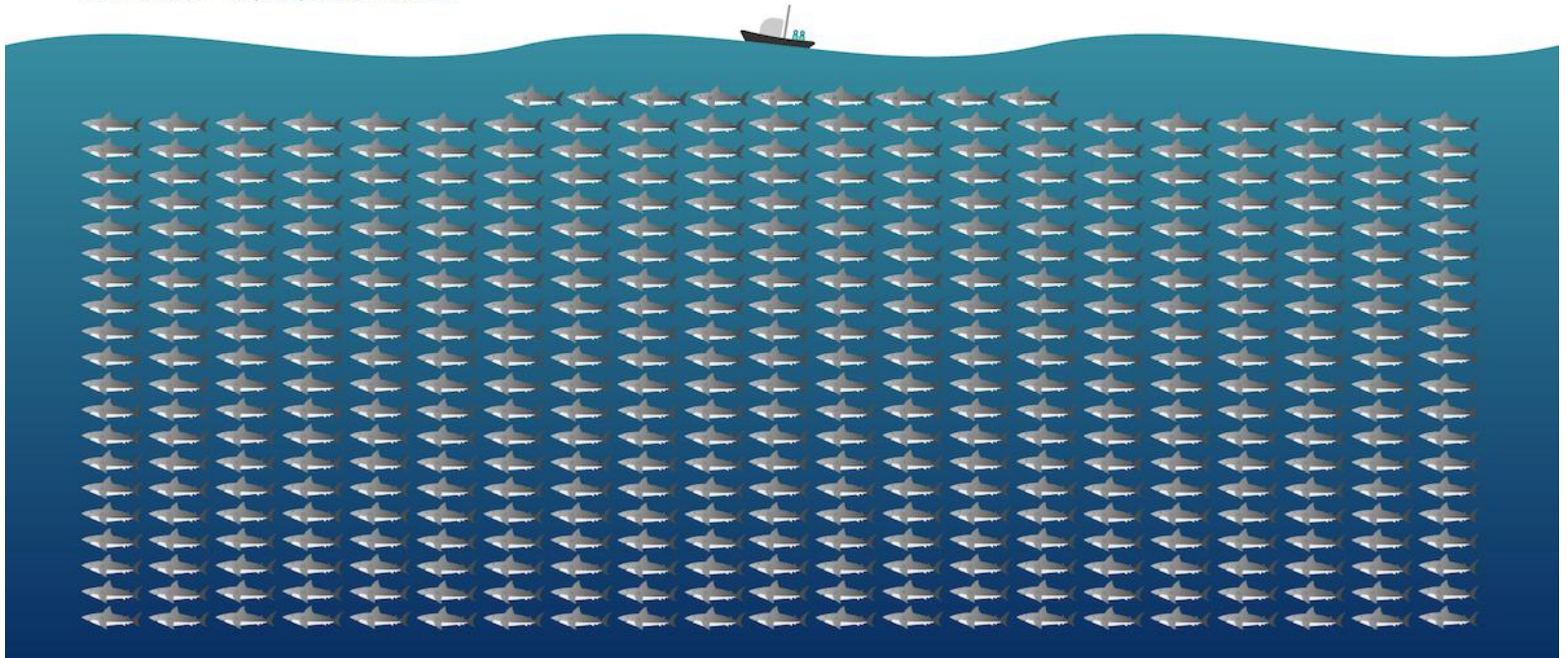


	Giải trình tự theo kiểu cũ		Giải trình tự thế hệ mới (NGS)	
Platform	Sanger		NovaSeq X Plus	
Số bộ gen người	12		128	
Thời gian	~10 năm (1990 đến 2000)		~ 48 h	
Giá thành	~3 tỉ USD		~200 USD	
Chất lượng	Không xác định		Q30 > 85%	
Chứng nhận	Không xác định		IVD	

How big is 40 exabytes?

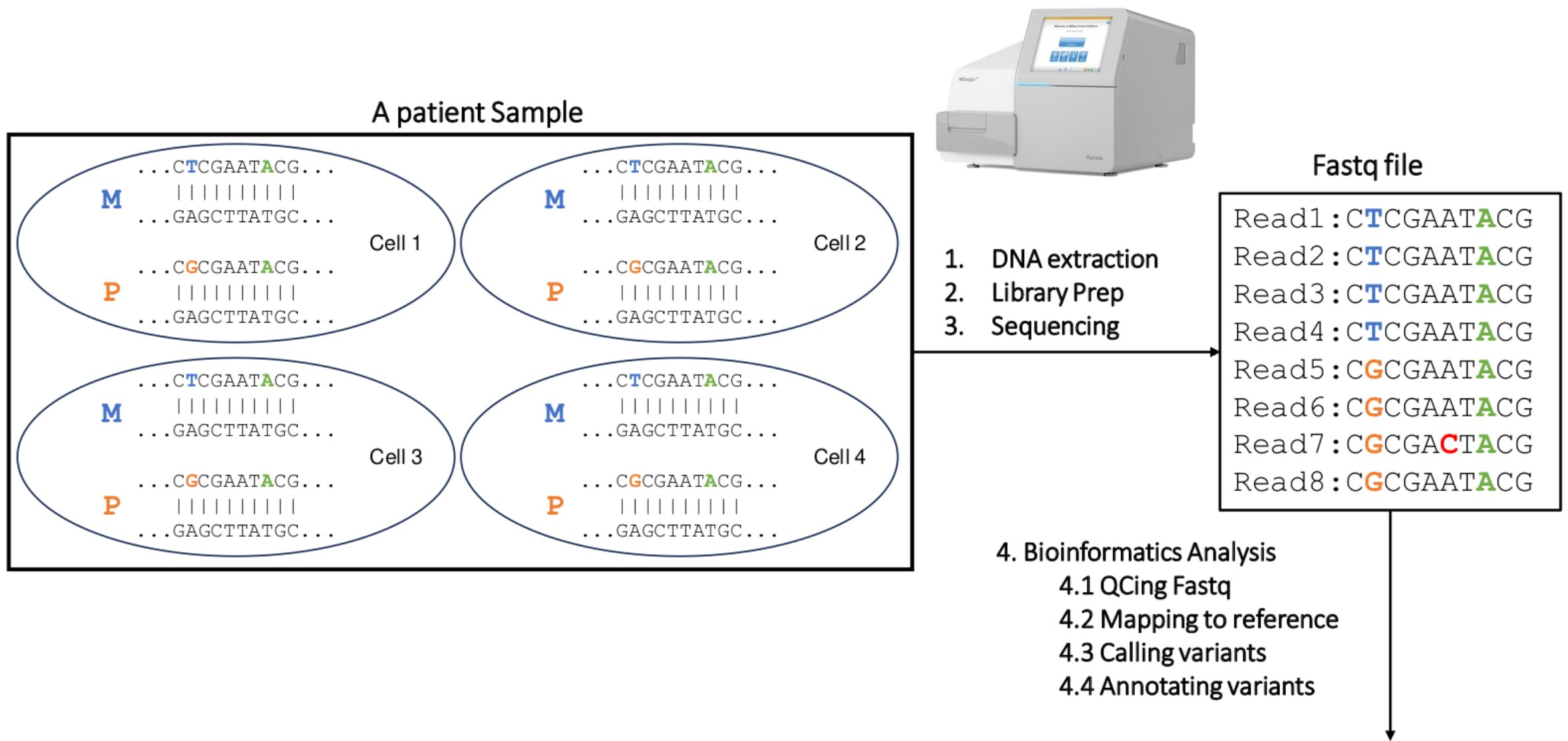
Genomics projects will generate 40 exabytes of data in the next decade.

Each shark = 100,000,000 GB of data



Quy trình **XÉT NGHIỆM** gen bằng phương pháp giải trình tự thế hệ mới (NGS)

Các bước trong XÉT NGHIỆM gen bằng phương pháp giải trình tự thế hệ mới (1)



Các bước trong XÉT NGHIỆM gen bằng phương pháp giải trình tự thế hệ mới (2)

4.2 Mapping reads to reference

Reference: ...CTCGAATGCG...		
Read1:	CTCGAATACG	Maternal
Read2:	CTCGAATACG	
Read3:	CTCGAATACG	
Read4:	CTCGAATACG	
Read5:	CGCGAATACG	Paternal
Read6:	CGCGAATACG	
Read7:	CGCGAATACG	
Read8:	CGCGAATACG	

Heterozygous

Homozygous

4.3 Calling variants

ANN=G|stop_gained|HIGH|OR4F5|ENSG00000186092|transcript|ENST0000641515.2|protein_coding|3/3|c.822T>G|p.Trp274*|882/2618|822/981|274/326||Pathogenic

ANN=A|frameshift_variant|HIGH|ZSWIM2|ENSG00000163012|transcript|ENST00000295131.3|protein_coding|9/9|c.1238G>A|p.Ile413|1293/2451|1238/1902|413/633||;LOF=(ZSWIM2|ENSG00000163012|1|1.00)

4.4 Annotating variants

```
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##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=AD,Number=2,Type=Integer,Description="Read depth for each allele">
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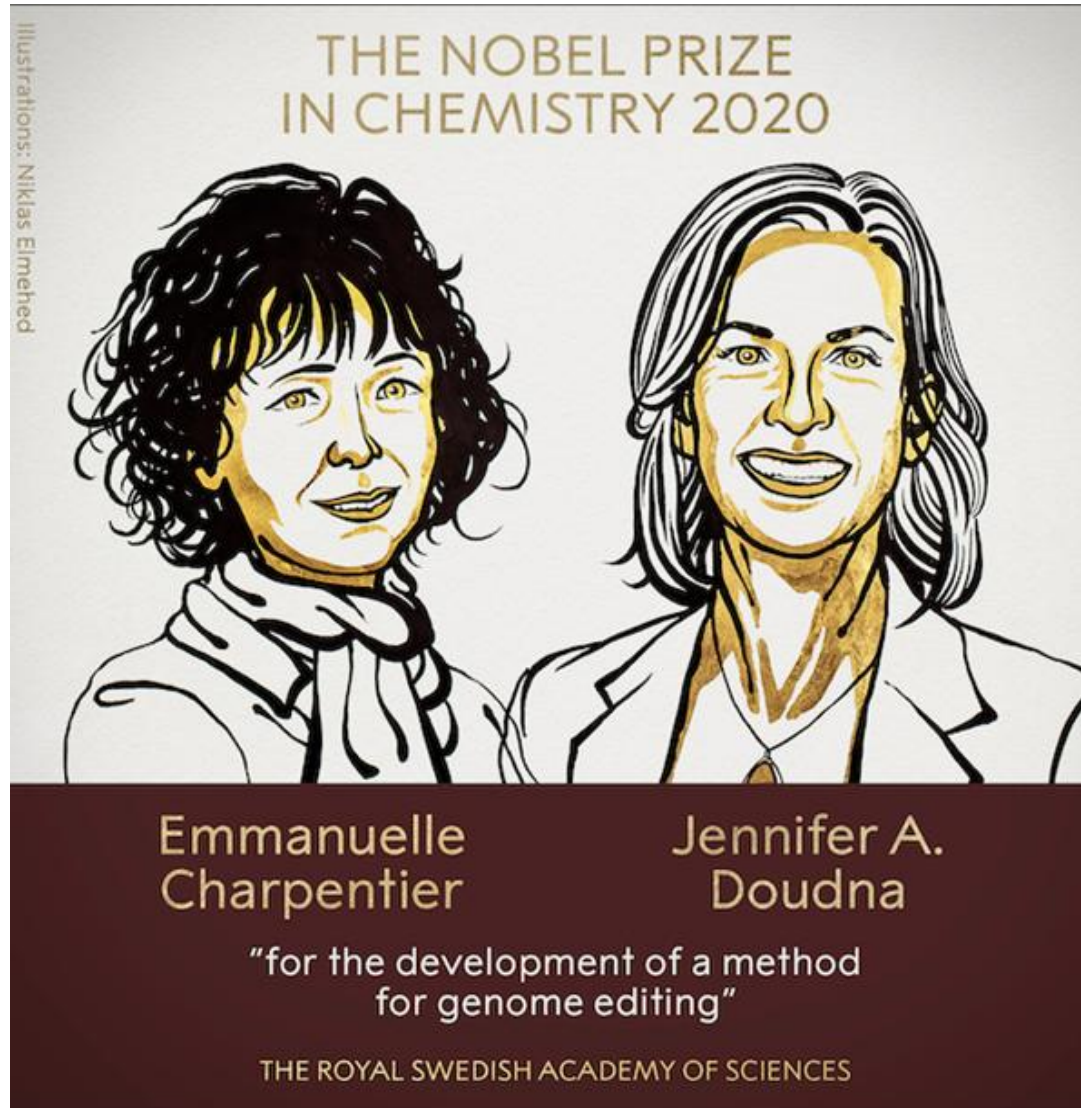
#CHROM	POS	ID	REF	ALT	QUAL	FILTER	FORMAT	Sample1
20	14370	rs6054257	T	G	129	PASS	GT:GQ:DP:AD	0/1:48:8:4,4
20	17330	.	G	A	150	PASS	GT:GQ:DP:AD	1/1:49:8:8,8

Hội đồng Hệ Gen: phiên giải và hội chẩn những biến thể trong báo cáo kết quả NGS



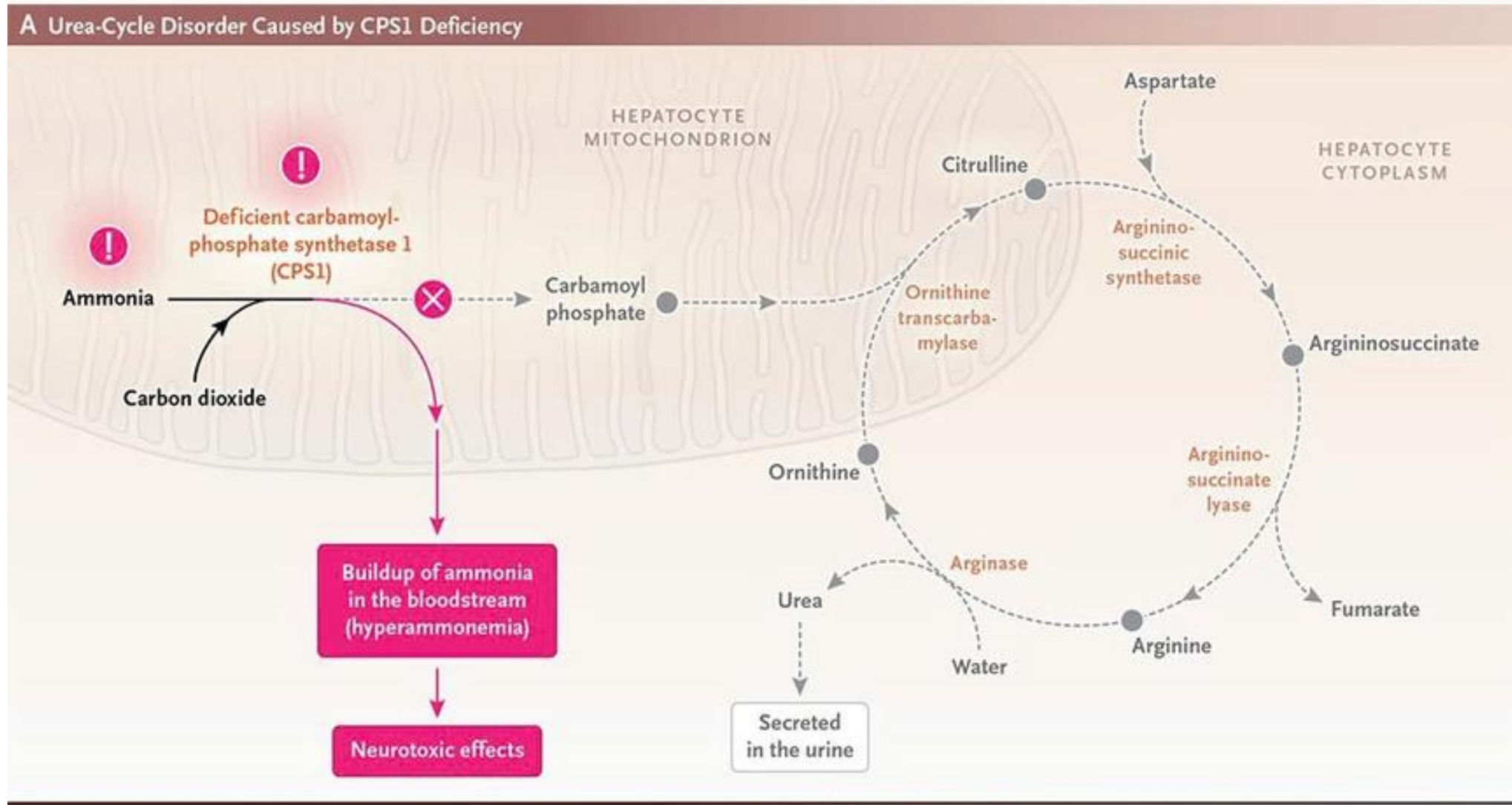
<https://ggba.swiss/en/the-first-medical-genomics-center-opens-in-geneva/>

LIỆU CÓ GIÚP ÍCH?



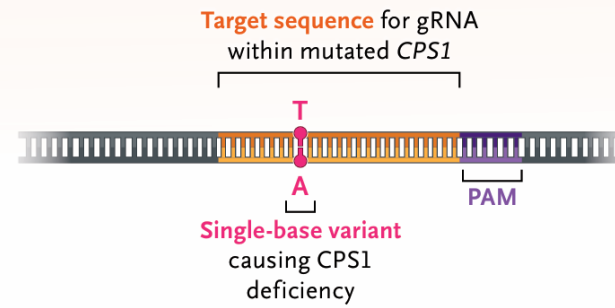
CRISPR-CAS9

New England Journal of Medicine (NEJM), May 2025

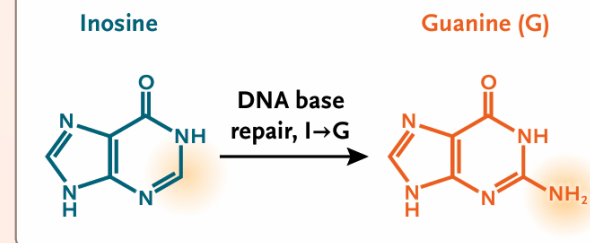
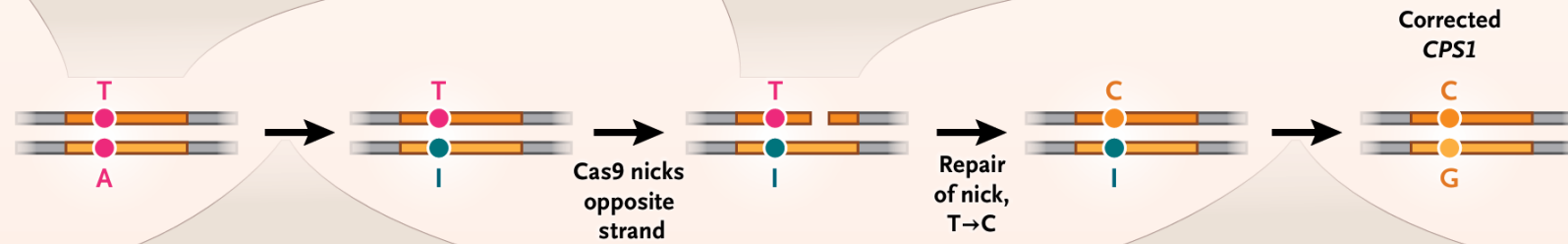
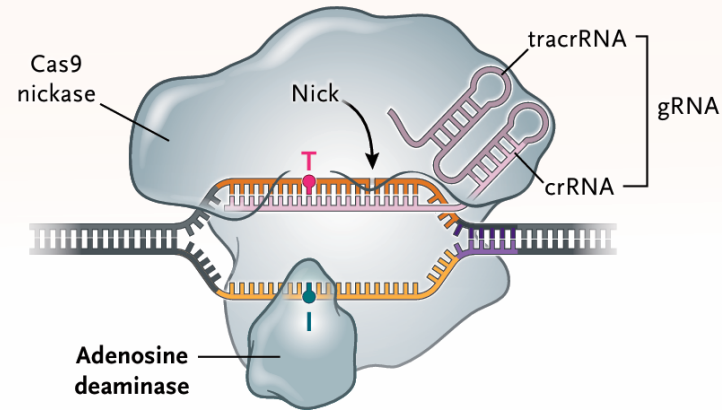


B Adenine Base Editor–Mediated Repair of *CPS1* Variant

CPS1 with Pathogenic Variant



K-abe Base Editor Bound to Target Sequence of *CPS1*



Gene-editing therapy made in just 6 months helps baby with life-threatening disease

Custom CRISPR paves the way for treating genetic disorders in tailormade ways

May 2025

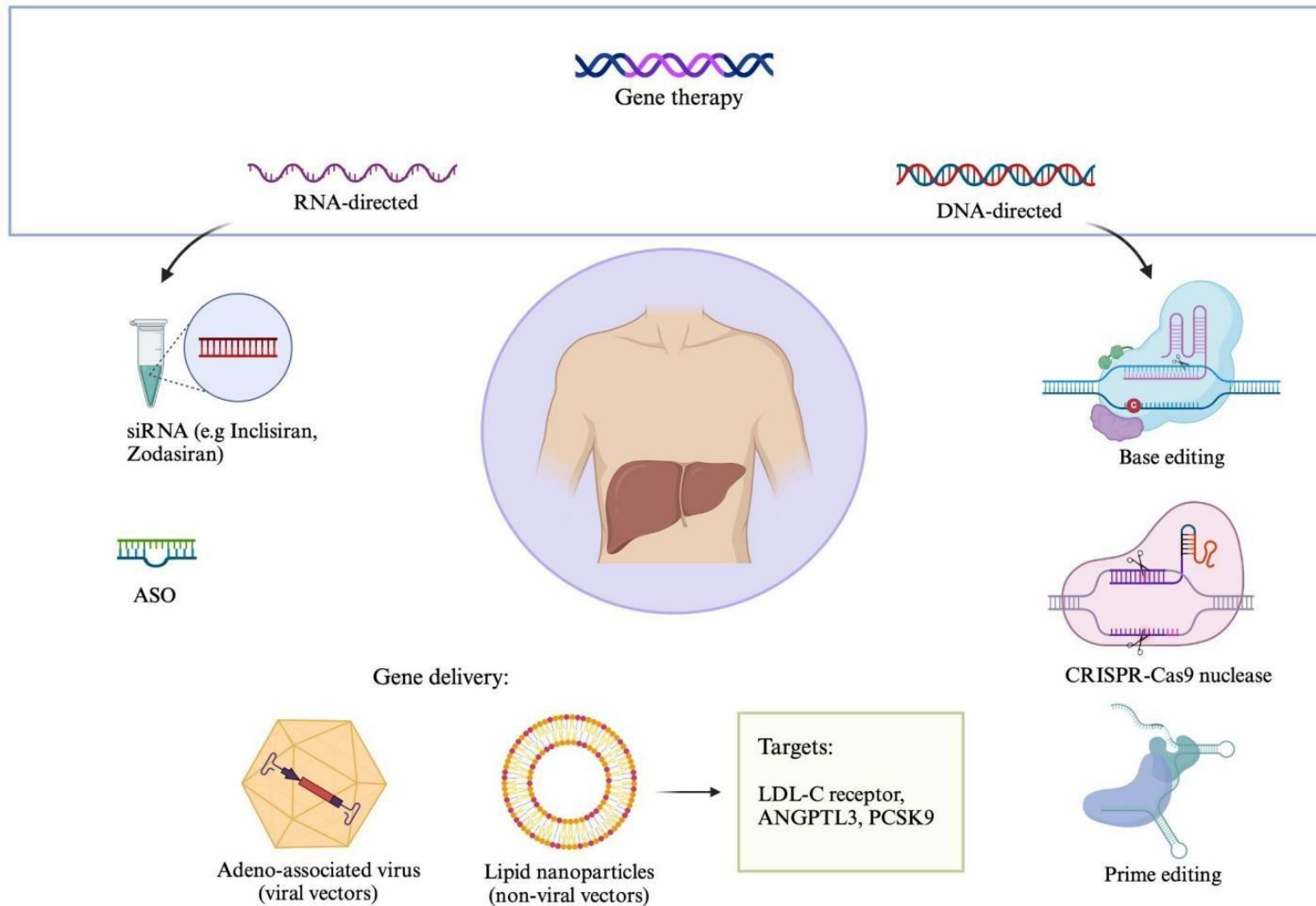
13 MAY 2023 · 1:00 PM ET · BY JOCELYN KAISER



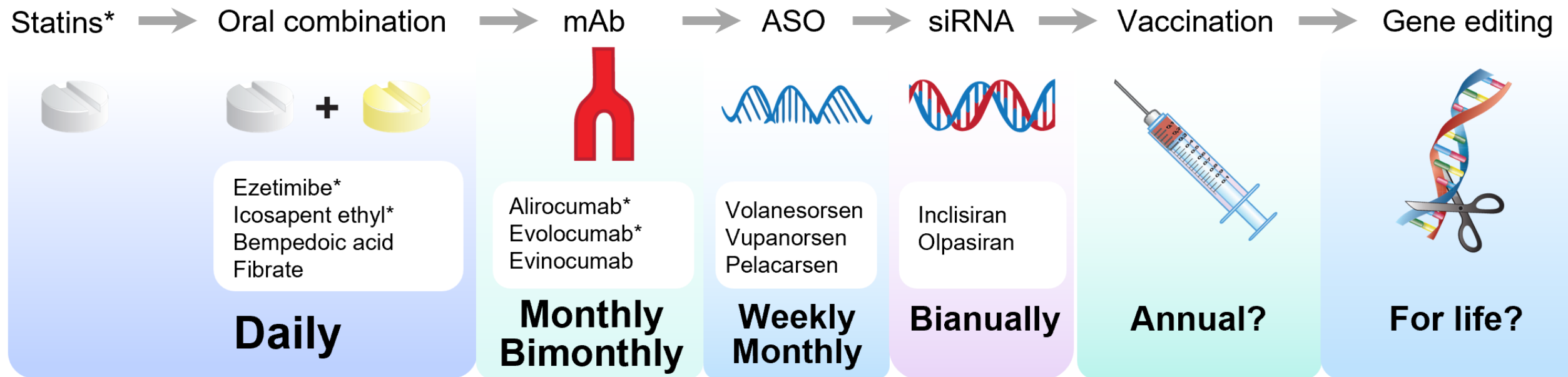
An infant who received a customized gene-editing treatment, shown here with researchers Kiran Musunuru (left) and Rebecca Ahrens-Nicklas, now needs less medicine to defuse a blood buildup of ammonia. . From "Gene-editing therapy made in just 6 months helps baby with life-threatening disease" by Jocelyn Kaiser, 2023. <https://www.science.org/content/article/gene-editing-therapy-made-just-6-months-helps-baby-life-threatening-disease>

Tăng cholesterol máu gia đình (Familial Hypercholesterolemia - FH)

Bảng gen cho bệnh tăng cholesterol máu gia đình: APOB, LDLR, LDLRAP1 và PCSK9



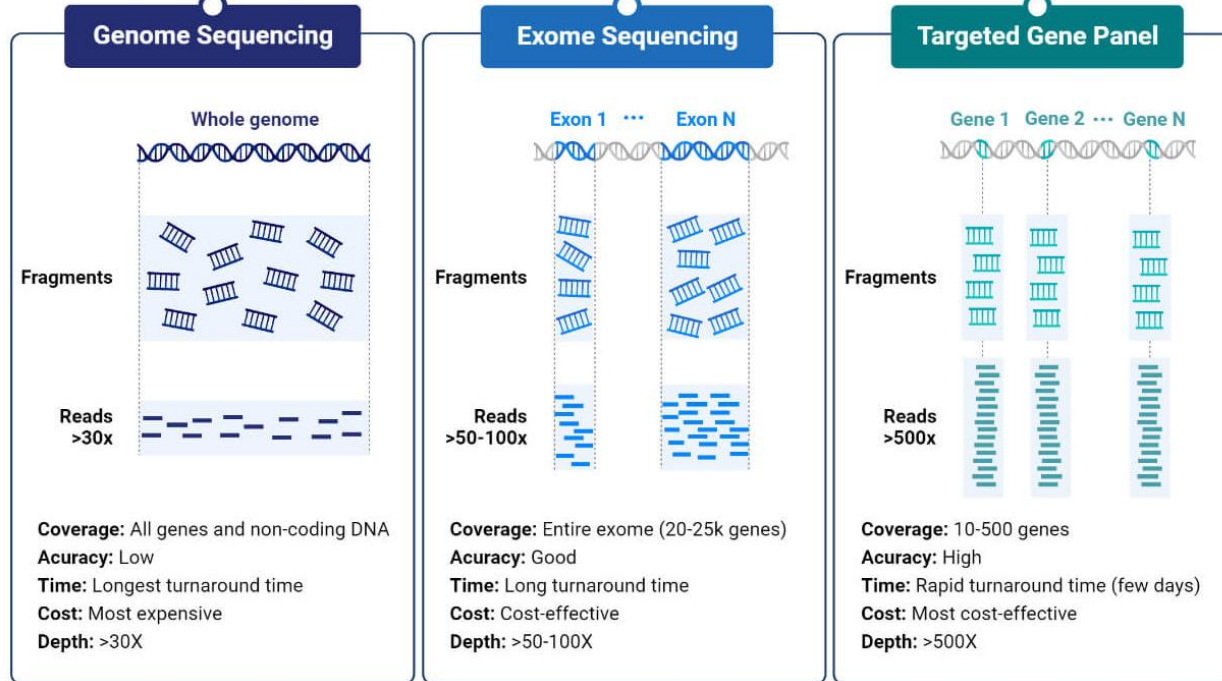
Điều trị tăng cholesterol máu gia đình (Familial Hypercholesterolemia - FH)



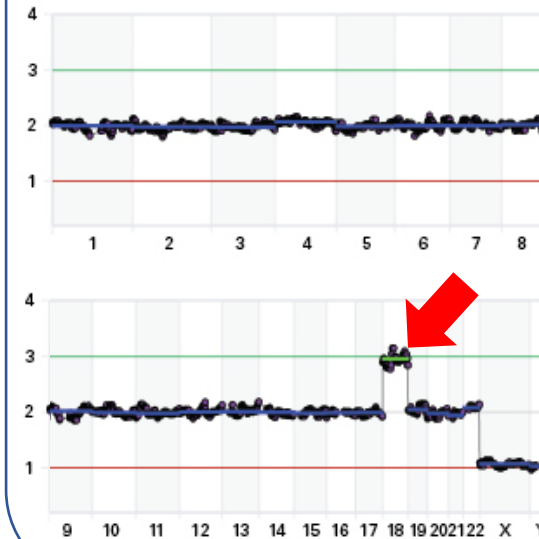
Từ slide GS TS. Trương Quang Bình - ĐHYD TP Hồ Chí Minh

Ứng dụng giải trình tự gen thế hệ mới trong lâm sàng

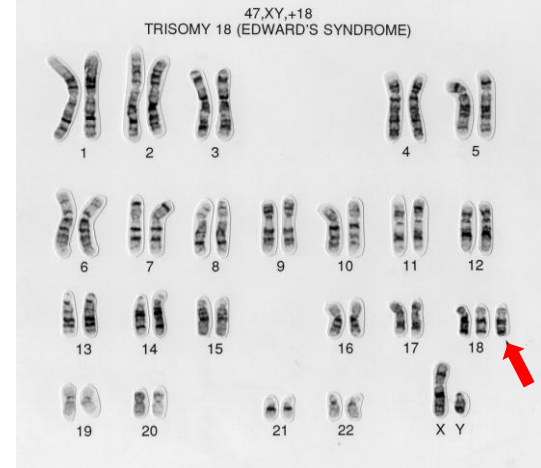
Single Nucleotide Variant (SNV)



Copy Number Variation (CNV) Low-Pass Genome Sequencing



Karyotype



<https://microbenotes.com/next-generation-sequencing-ngs/>

<https://wellcomecollection.org/search/images?query=eaahtz2u#>

Năm ví dụ về **XÉT NGHIỆM** gen
cho **BỆNH DI TRUYỀN** và **UNG THƯ**

Ví dụ 1: Phát hiện sớm bệnh động kinh bằng xét nghiệm gen (1)

Whole Exome Sequencing - WES

IGV - DEE001 – **CDKL5**



Ví dụ 1: Phát hiện sớm bệnh động kinh bằng xét nghiệm gen (2)

Whole Exome Sequencing - WES

DEE001							
Gene	Amino acid change	cDNA	Variant type	Allele frequency	Transcript	Variant effect	ClinVar significance
CPT2	p.Arg631Cys	c.1891C>T	SNP	0.5	ENST00000371486.4	Missense variant	Pathogenic
CDKL5	p.Gln881Ter	c.2641C>T	SNP	0.5	ENST00000623535.2	Stop gained (Nonsense)	Pathogenic
GALC		c.1162-4del	DEL (1bp)	1	ENST00000261304.7	Intron variant	Conflicting interpretations of pathogenicity
TUBB2B	p.Ala248Val	c.743C>T	SNP	0.5	ENST00000259818.8	Missense variant	Conflicting interpretations of pathogenicity

DEE001 – **CDKL5** (Xp22.13)

rs1057519541

Current Build 156
Released September 21, 2022

Organism	Homo sapiens	Clinical Significance	Reported in ClinVar
Position	chrX:18628515 (GRCh38.p14)	Gene : Consequence	CDKL5 : Stop Gained
Alleles	C>T	Publications	1 citation
Variation Type	SNV Single Nucleotide Variation	Genomic View	See rs on genome
Frequency	None		

Variant Details
Clinical Significance
HGVS
Submissions
History
Publications
Flanks

Allele: T (allele ID: 362353)

ClinVar Accession	Disease Names	Clinical Significance
RCV000416943.1	Focal epilepsy	Pathogenic

https://www.ncbi.nlm.nih.gov/snp/rs1057519541#clinical_significance

NGUYEN Thuy-Minh-Thu, MD
 Nguyen Le Duc Minh, MD

Ví dụ 2: Sàng lọc bệnh ung thư bằng xét nghiệm gen

Gene Panel – 53 genes của MGI

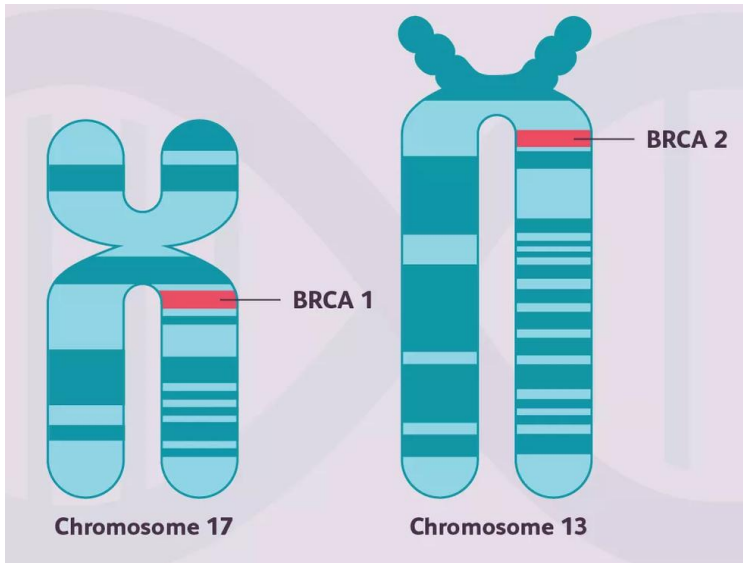
Gene	Amino Acid Change	Coding	Variant Type	Allele Frequency	Transcript	Variant effect	ClinVar Significance
CTNNB1	p.Ser33Tyr	c.98C>A	SNP	0.5	ENST00000349496	MISSENSE	Pathogenic/ Likely_pathogenic
PIK3CA	p.Gly914Arg	c.2740G>A	SNP	0.5	ENST00000263967	MISSENSE	Pathogenic
KRAS	p.Gly12Asp	c.35G>A	SNP	0.5	ENST00000256078	MISSENSE	Pathogenic
BRCA2	p.Ile2675AspfsTer6	c.8021dup	INS	0.5	ENST00000544455	FRAMESHIFT	Pathogenic

Nguyen Le Duc Minh, MD

Ví dụ 3: Hỗ trợ điều trị bệnh ung thư vú bằng xét nghiệm gen

Gene Panel BRCA1 và BRCA2

Olaparib (AstraZeneca) là một loại thuốc dùng để duy trì điều trị ung thư vú, buồng trứng, tuyến tiền liệt và tuyến tụy giai đoạn tiến triển có đột biến BRCA ở người lớn.



KẾT QUẢ XÉT NGHIỆM *BRCA1/2*

Họ và tên : TRẦN THỊ X.	Tuổi : 1956	Giới tính : NỮ
Số hồ sơ:		
KHOA:	BS điều trị:	
Bệnh phẩm : Mô vú ứ sữa	Số block: XXXX	
Yêu cầu: Xét nghiệm giải trình tự gen trên hệ thống Miseq [02 gen <i>BRCA1</i> và <i>BRCA2</i>]		
Ngày nhận chỉ định: 20/10/2022		Ngày thực hiện: 25/10/2022

Chẩn đoán lâm sàng: Ung thư buồng trứng dịch trong grade cao/ Ung thư vú trái

Chất lượng mẫu: **MẪU ĐẠT** (kích thước 17mm x 16mm, thành phần bướu 70%)

Phương pháp: Giải trình tự gen bằng phương pháp NGS cho 02 gen *BRCA1* và *BRCA2*

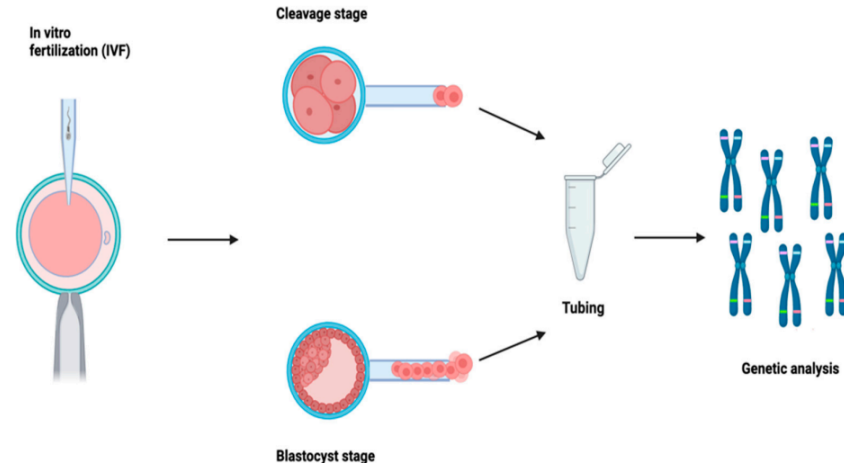
- Hệ thống xét nghiệm Illumina MiSeqDx (CE/US-IVD)
- Bộ xét nghiệm: NGeneBio BRCAaccuTest™ Plus (CE-IVD)
- Phần mềm hỗ trợ phân tích kết quả: NGeneBio NGeneAnalySys™ (CE-IVD)

Kết quả: **PHÁT HIỆN 1 BIẾN THỂ MẤT ĐOẠN NUCLEOTIDE NHỎ (DEL) GÂY BỆNH TRÊN GEN *BRCA1***

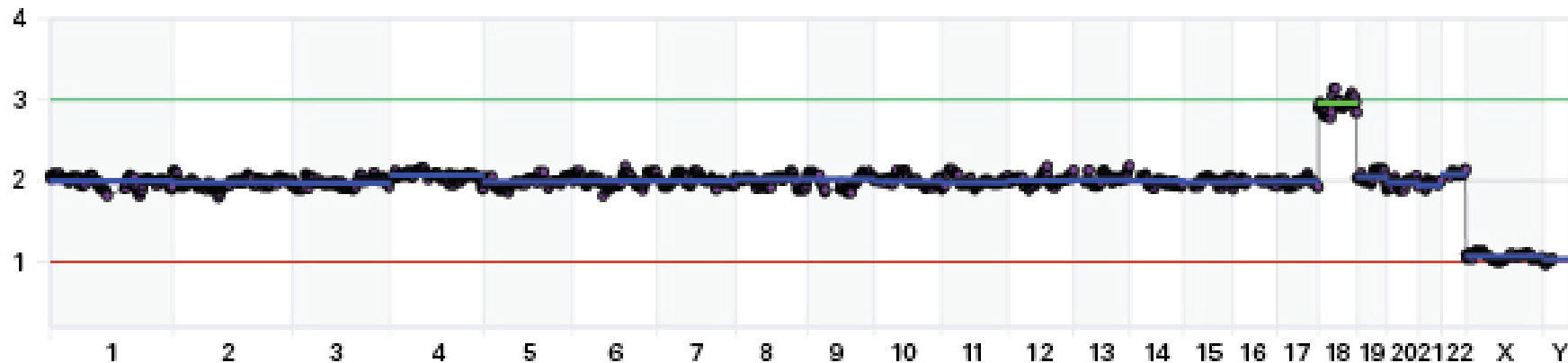
MÔ TẢ KẾT QUẢ			
Gen	Biến thể gây bệnh/có khả năng gây bệnh	Tỷ lệ	Phân loại
<i>BRCA1</i>	c.5335del (p.Gln1779AsnfsTer14)	82.03%	Gây bệnh (Pathogenic)
<i>BRCA2</i>	Không phát hiện	Không	Không

Ví dụ 4: Sàng lọc phôi trong hỗ trợ sinh sản IVF

Xét nghiệm tiền làm tử PGT-A

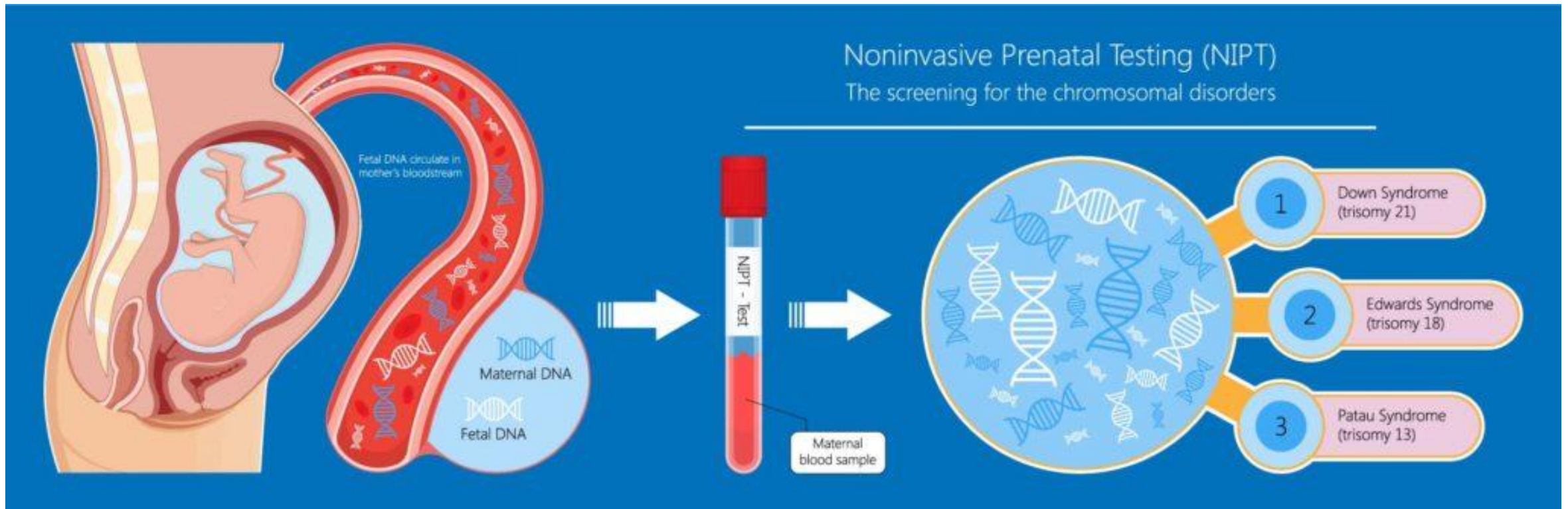


10.3390/genes14112095

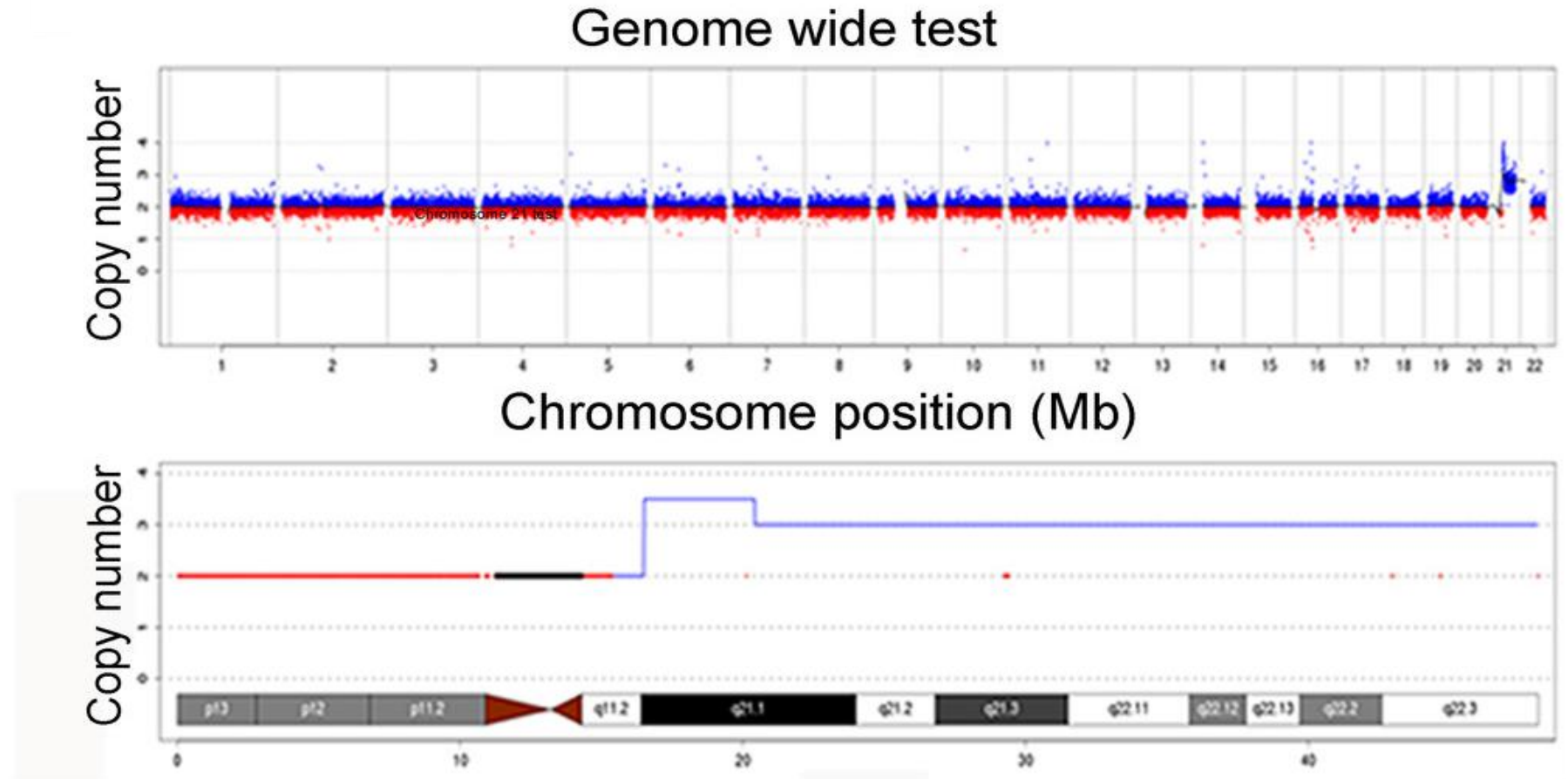


Mẫu phôi có 3 nhiễm sắc thể 18 (trisomy) trong DNA hệ gen

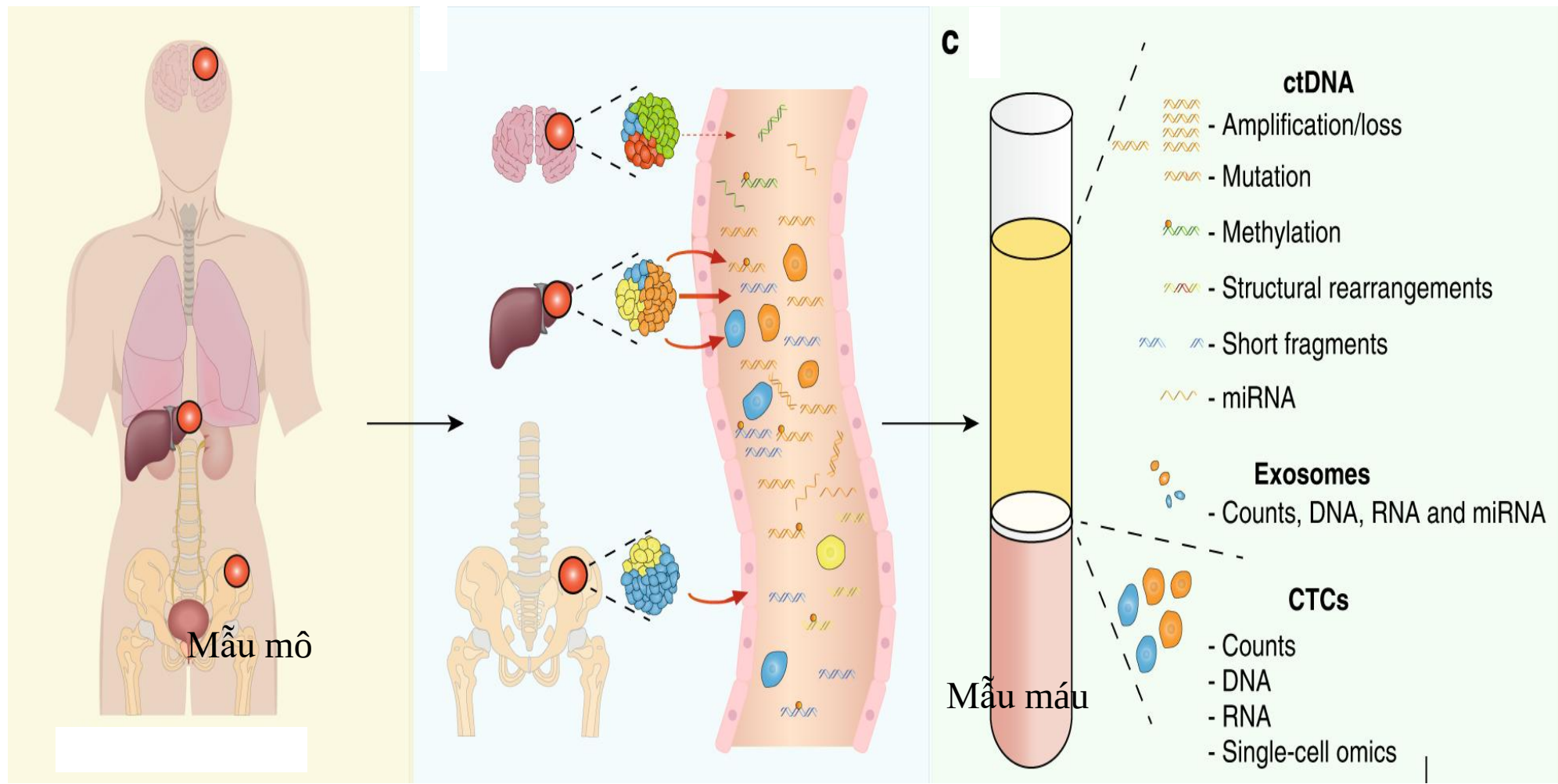
Ví dụ 5: Sàng lọc bất thường nhiễm sắc thể ở thai nhi



Kết quả NIPT: Bất thường nhiễm sắc thể 21 ở thai nhi

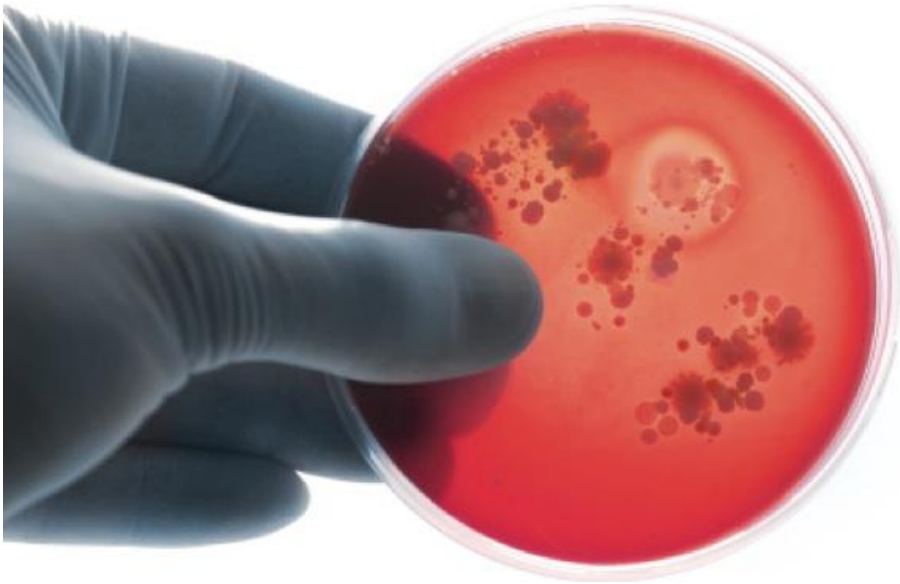


Mẫu cho XÉT NGHIỆM gen: mẫu mô và máu

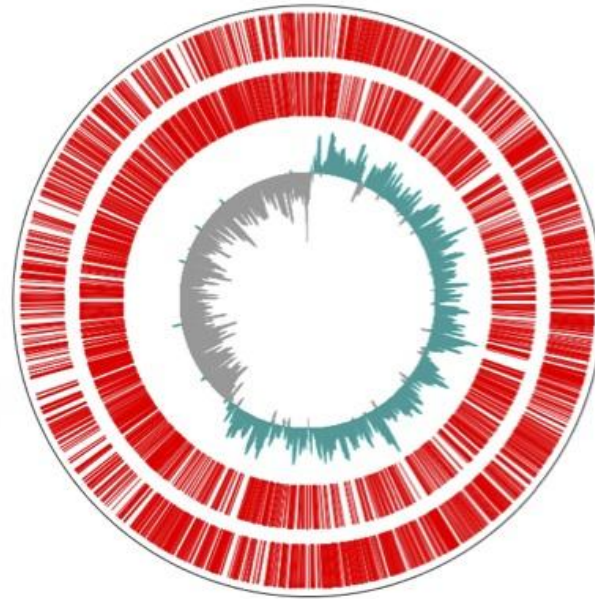


Ứng dụng NGS trong bệnh nhiễm

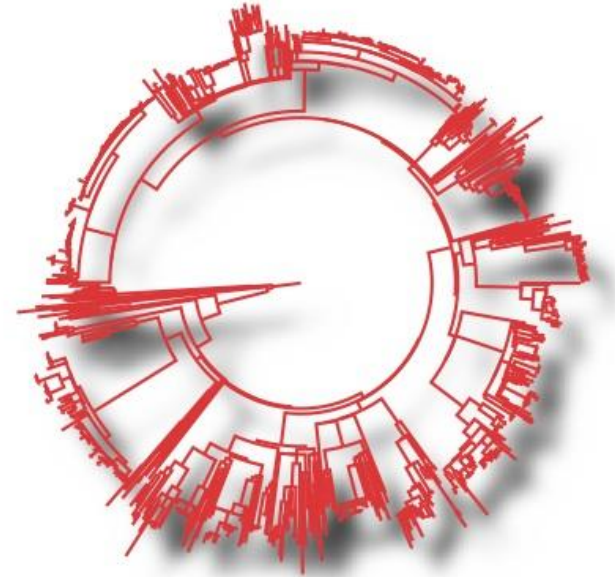
Phân tích bộ gen vi khuẩn: nghiên cứu một tác nhân gây bệnh ở mỗi mẫu phân lập



Phân lập



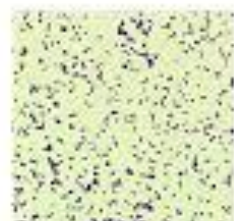
Giải trình tự bộ gen



Serotyping/genotyping

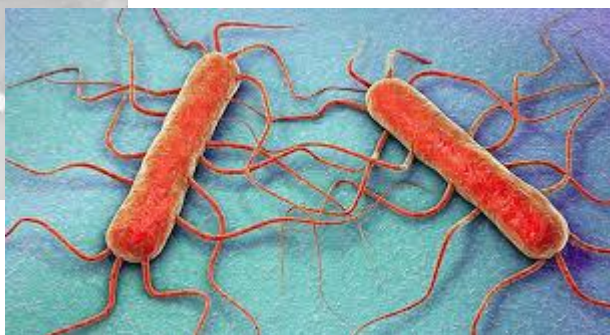
Listeria monocytogenes

microbeonline



Gram positive coccobacilli

Beta-hemolytic colonies



Listeria monocytogenes EGD-e, complete genome

GenBank: AL591824.1

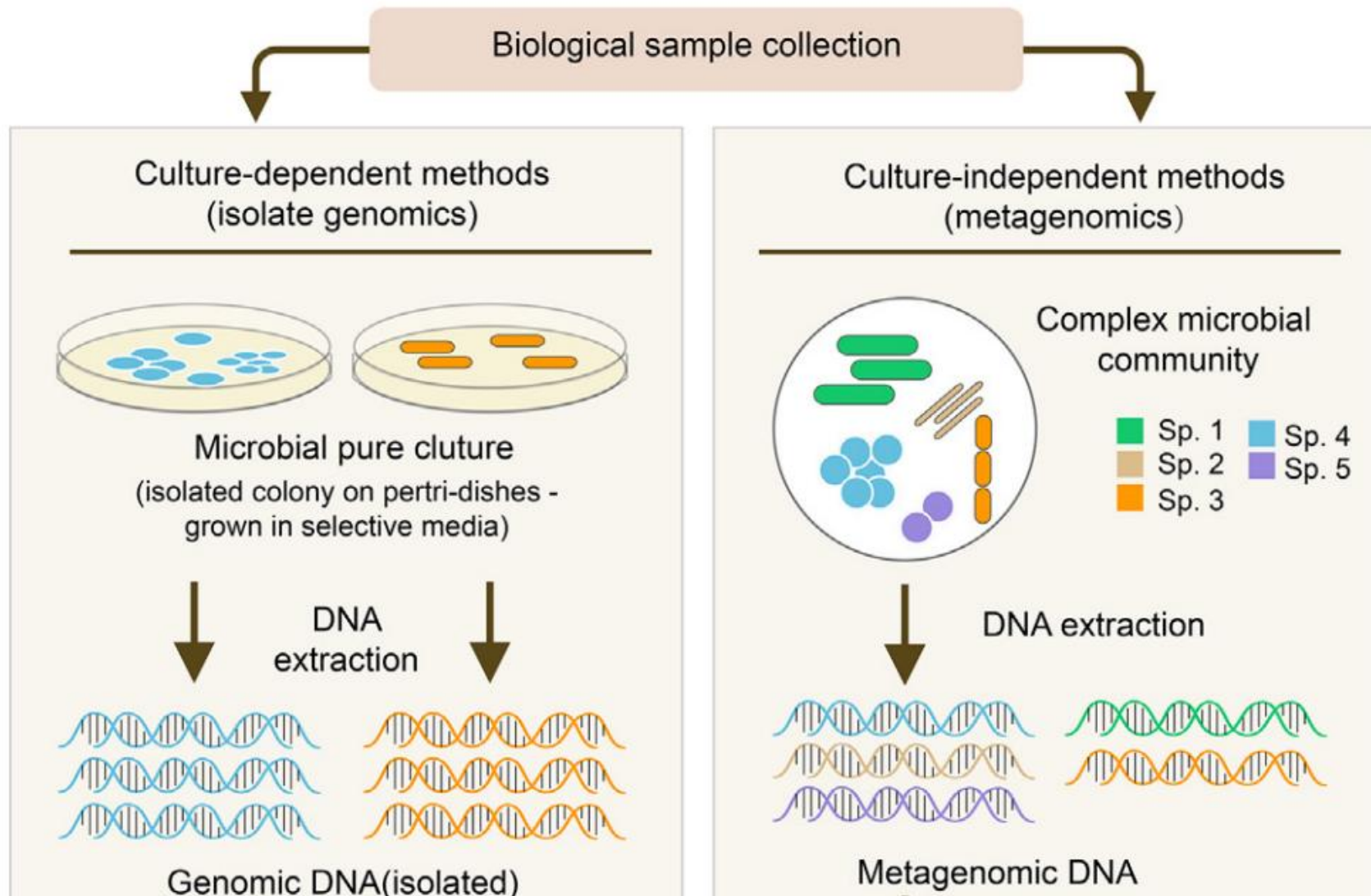
[GenBank](#) [Graphics](#)

>AL591824.1 Listeria monocytogenes EGD-e, complete genome

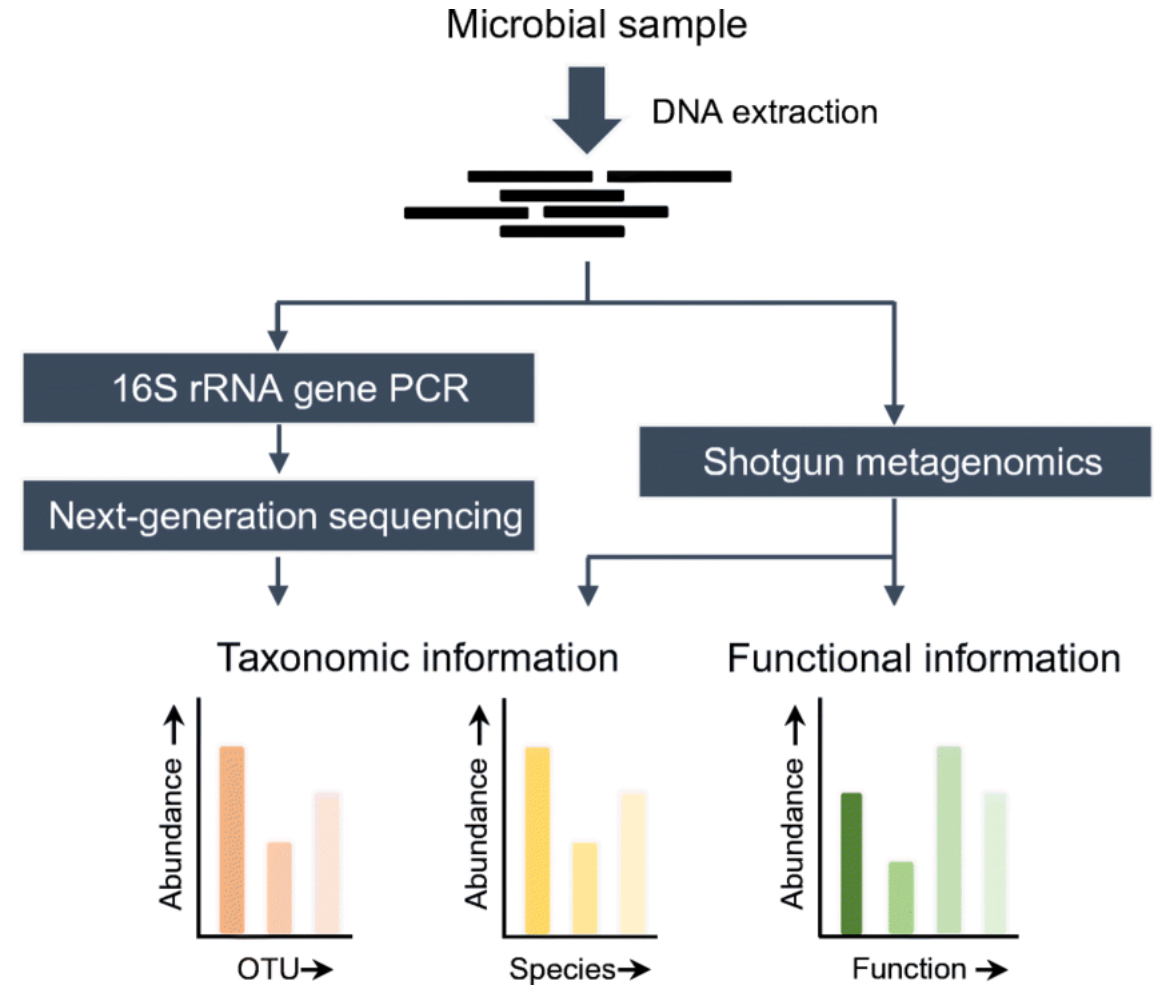
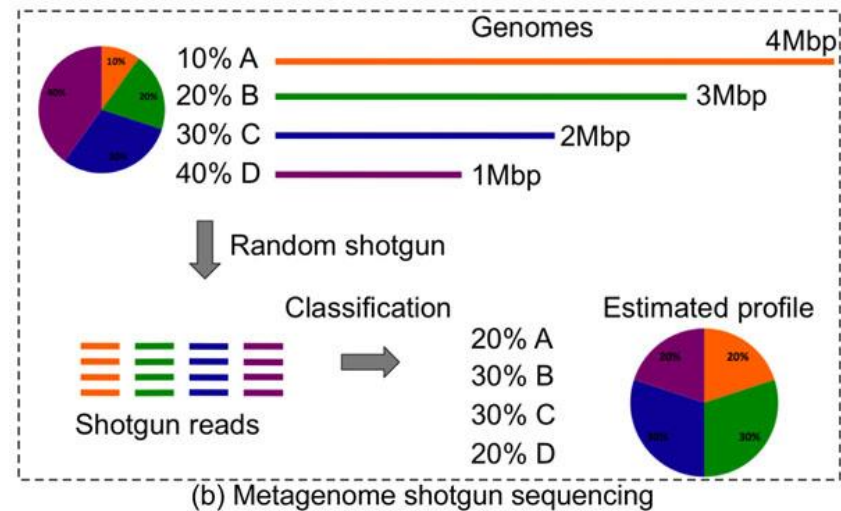
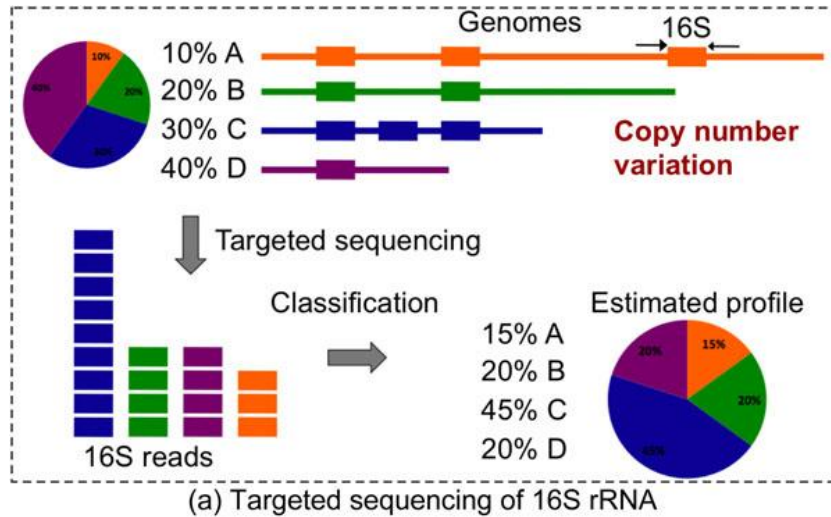
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<https://www.ncbi.nlm.nih.gov/nuccore/AL591824.1?report=fasta>

Phân tích một bộ gen vs nhiều bộ gen



Target vs Shotgun Microbiome Analysis



Precision microbiome testing: STI, HPV and AMR

Women's Health Test

Page 1 of 4

Patient Name:	nan	Provider:	Mony Sary, Yap Chew, Wendy Ullmer	Patient ID:	VS8
Gender:	nan	Provider NPI:	nan	Specimen ID:	KH20.45474
DOB:	nan	Order Date:	nan	Specimen Type:	nan
Age:	nan	Health Status Reported:	nan	Collection Date:	nan

Sexually Transmitted Infections

Name	Associated Condition	Result
<i>Neisseria gonorrhoeae</i>	Gonorrhea, urethritis, pelvic inflammatory disease, gonococemia, gonococcal ophthalmia neonatorum	Detected
<i>Chlamydia trachomatis</i>	Chlamydia, cervicitis, urethritis, pelvic inflammatory disease	Not Detected
<i>Mycoplasma genitalium</i>	Urethritis, cervicitis, pelvic inflammatory disease	Not Detected
<i>Treponema pallidum</i>	Syphilis	Not Detected
<i>Haemophilus ducreyi</i>	Chancroid	Not Detected
<i>Trichomonas vaginalis</i>	Trichomoniasis	Not Detected
<i>Human papillomavirus</i>	Cervical and anogenital cancers, genital warts	Detected
<i>Herpes simplex virus</i>	Genital herpes, oral herpes	Not Detected

Viruses Detected

Name	Associated Condition
<i>Human papillomavirus 62 (HPV 62)</i>	Unknown risk for cervical cancer

Note: Human papillomavirus (HPV) 16, 18, 31, 33, 35, 39, 45, 51, 56, 58, 59, and 68 are considered high-risk or probable high-risk due to their association with cervical cancer. HPV 6, 11, 42, 43, and 44 are considered low-risk for cervical cancer but may cause genital warts. Other HPV genotypes found in the sample may have intermediate or unknown risk for cervical cancer.

Antimicrobial Resistance Genes Detected

AMR Gene Name	Function	Drug Class
<i>Neisseria.gonorrhoeae.folP</i>	Dihydropteroate synthase (mutated)	Sulfonamide

Women's Health Test

Page 1 of 4

Patient Name:	nan	Provider:	Yap Chew	Patient ID:	nan
Gender:	nan	Provider NPI:	nan	Specimen ID:	202122865
DOB:	nan	Order Date:	nan	Specimen Type:	nan
Age:	nan	Health Status Reported:	nan	Collection Date:	nan

Sexually Transmitted Infections

Name	Associated Condition	Result
<i>Neisseria gonorrhoeae</i>	Gonorrhea, urethritis, pelvic inflammatory disease, gonococemia, gonococcal ophthalmia neonatorum	Not Detected
<i>Chlamydia trachomatis</i>	Chlamydia, cervicitis, urethritis, pelvic inflammatory disease	Not Detected
<i>Mycoplasma genitalium</i>	Urethritis, cervicitis, pelvic inflammatory disease	Not Detected
<i>Treponema pallidum</i>	Syphilis	Not Detected
<i>Haemophilus ducreyi</i>	Chancroid	Not Detected
<i>Trichomonas vaginalis</i>	Trichomoniasis	Not Detected
<i>Human papillomavirus</i>	Cervical and anogenital cancers, genital warts	Detected
<i>Herpes simplex virus</i>	Genital herpes, oral herpes	Not Detected

Viruses Detected

Name	Associated Condition
<i>Human papillomavirus 52 (HPV 52)</i>	High-risk for cervical cancer
<i>Human papillomavirus 68 (HPV 68)</i>	High-risk for cervical cancer

Note: Human papillomavirus (HPV) 16, 18, 31, 33, 35, 39, 45, 51, 56, 58, 59, and 68 are considered high-risk or probable high-risk due to their association with cervical cancer. HPV 6, 11, 42, 43, and 44 are considered low-risk for cervical cancer but may cause genital warts. Other HPV genotypes found in the sample may have intermediate or unknown risk for cervical cancer.

Third Generation Sequencing – PacBio Long Read

<https://www.youtube.com/watch?v=7yYPHatccgw>

**Xin chân thành cảm ơn
sự lắng nghe của quý khách mời!**

Luu Phuc Loi, PhD

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